

Factorizable Rational Quantum Channels at an Explicit Distance from Finite Tracial Baths

Nidhal Mghirbi

Seth Douglas

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Abstract

A quantum channel can always be implemented by a unitary interaction with an environment that starts in a pure state. With a finite, maximally mixed environment, the implementable channels are the noisy operations. A tracial von Neumann algebra as environment gives the factorizable channels. By a theorem of Haagerup and Musat, combined with $\text{MIP}^* = \text{RE}$, some of these lie outside the closure of the noisy operations. In earlier work we wrote down explicit rational channels of this kind, with no value for their distance from the closure. Using Wang and Zhi's dimension-uniform reflection estimate, we construct a factorizable unital channel on twelve qubits with Kraus entries in $\{0, 1, \pm 3/5, 4/5\}$. Its normalized Choi and diamond distances from the closure are at least $2^{-(2\mathfrak{J}+50)}$, with $\mathfrak{J} = 2^{1122000}$. The bound comes from a compiler theorem: group relations that are rigid uniformly in the matrix dimension give a rational channel with an explicit distance bound. Unlike the bound of our earlier work, it needs reference positions only for the identity, the generators and their inverses. For the earlier channels, a new spectral certificate for the root groups of their presentation, with Wang and Zhi's estimates, gives an explicit tolerance. The fifteen-qubit channel and its two variants are then at distance at least $2^{-(2\mathfrak{J}+41)}$ in both metrics. By our earlier bound, the fourth channel, with a reference position for every group element it uses, is at Choi distance at least $2^{-(2\mathfrak{J}+36)}$ and diamond distance at least $2^{-(2\mathfrak{J}+21)}$. These numbers are tiny because Wang and Zhi's tolerance is $2^{-\mathfrak{J}}$. Their method builds on the group construction of Kun and Thom and on nonhyperlinearity results of Thom, Liu, and Alekseev, Liu and Thom. The finite certificates are checked by exact computation.

1 Introduction

1.1 The question

Stinespring's theorem implements every quantum channel by a unitary interaction with an environment that starts in a pure state [26]. Suppose instead that the environment must start maximally mixed, so that it supplies no purity. The channels implementable this way, with a finite environment, are the *noisy operations* of Horodecki, Horodecki and Oppenheim [10]. Which channels are limits of noisy operations?

Every noisy operation is *factorizable*: it factors through a finite von Neumann algebra with a faithful normal tracial state [3, Definition 6.2], [8, Theorem 2.2]. The factorizable channels form a closed set [8, Remark 1.4(b) and proof of Theorem 6.1]. So every limit of noisy operations is unital and factorizable. The ultraproduct argument in the proof of [9, Theorem 3.6] shows more: such a limit factors through an ultrapower of the hyperfinite II_1 factor. Haagerup and Musat showed that the converse, for all channels on all matrix algebras, is equivalent to a positive answer to Connes' embedding problem [9, Theorem 3.7]. Since $\text{MIP}^* = \text{RE}$ [11], the converse fails: some factorizable channels are not limits of noisy operations. That argument does not produce a channel one can write

down. The standard route from a non-embeddable algebra to such a channel goes through Schur multipliers and Kirchberg’s unitary correlation matrices, in the form of Dykema and Juschenko [12], [6], [9, proof of Theorem 3.7], [18, Corollary 3.11]. Musat and Rørdam showed that the limits of noisy operations are exactly the channels given by hyperlinear traces on the free product $M_d *_\mathbb{C} M_d$ [20, Proposition 3.2(ii) and Theorem 2.10]. Neither route gives a channel or a distance without an explicit non-embeddable input.

Explicit examples nearby lie on either side of the question. Nonfactorizable channels have no tracial realization at all [8]; Haagerup and Musat showed that every nonfactorizable unital channel violates the asymptotic quantum Birkhoff conjecture, and that such channels exist on M_n for every $n \geq 3$ [8, Theorem 6.1 and its proof]. For the Werner–Holevo channel W_3^- they computed the cb distance to the factorizable channels exactly: it is $4/27$ [9, Theorem 5.6(2)]. Channels that need an infinite-dimensional, indeed type II_1 , ancilla can still be approximated by noisy operations [19, Theorem 4.1]; Mori’s non-closure theorem for correlation matrices of unitaries [17, Theorem 2.3] lowers the dimension of such examples from 11 to 8 [18, Corollary 3.10]. The channels of this paper are factorizable and lie at an explicit distance from the limits of noisy operations.

1.2 Explicit channels

In earlier work [16] we wrote such channels down. We built a finite presentation of a nonhyperlinear group modelled on a Kun–Thom pair over $\mathbb{F}_5$ [14]. It has a witness word that every homomorphism into a tracial ultraproduct of matrix algebras must kill. This follows from Thom’s normalization theorem and his witness argument [27, Theorem 1.2 and Section 5.2], with the internality theorem of Liu [15, Theorem 1.2] in the form of Alekseev, Liu and Thom [1, Theorem B]. Rational 2×2 blocks turned the presentation into unital channels that are factorizable but not limits of noisy operations. Each basis position of such a channel lies either in a 2×2 block, which encodes one relation $xy = z$ among group elements, or in a *singleton* carrying one group element. The group elements that occur are the *labels*. There were four channels. The fully referenced channel Φ_{full} on M_{44086} (Φ_{ref} in [16]) has a singleton for every label. The reduced channels are Φ_1 on M_{31981} , with a singleton only for the identity; Φ_2 on M_{31982} , with singletons for the identity and the witness; and Φ_{15} , which pads Φ_2 to fifteen qubits with 786 further identity positions. The subscripts of Φ_1 and Φ_2 count singleton labels, not qubits. Their distances from the closure were shown to be positive, by an argument that rests on the preprints of Thom and of Alekseev, Liu and Thom, but no lower bound was given. A compactness argument gives a tolerance e_0 on the relator defects that forces the witness close to the identity, but no value for e_0 . The same work converted any value of e_0 into explicit bounds for Φ_{full} (Lemma 7.3 below). It noted that a quantitative support-leakage theorem would do the same for the reduced channels.

The group theory behind these constructions is recent. Building on Kun’s expander decomposition for Kazhdan groups [13, Theorem 1], OpenAI constructed the first nonsofic group [22]. Kun and Thom then showed that amalgamated doubles of suitable Kazhdan pairs are nonsofic [14, Theorems A and 4.1], and gave explicit pairs over every finite field [14, Theorem E]. Alekseev and Thom proved that centralizers of sofic approximations of Kazhdan groups are internal [2, Theorem 3.1], and asked for the matrix analogue [2, Open Problem 6.2(a)]. Thom showed that a positive answer makes these doubles nonhyperlinear, with an explicit witness commutator [27, Theorems 1.2 and 1.3 and Section 5.2]. Liu answered the problem positively [15, Theorem 1.2] and proved that a Kun–Thom wreath product is not hyperlinear [15, Theorem 1.1]. Alekseev, Liu and Thom proved a stronger internality theorem, for unitary tuples with spectral gap [1, Theorem B], and deduced that the doubles are not hyperlinear [1, Corollary C].

Forcing operator relations through channels or correlations has precedents. Musat and Rørdam

prescribe a correlation matrix of unitaries that forces projections to sum to an irrational multiple of the identity, and obtain a Schur-multiplier channel that needs a type II_1 ancilla [19, Theorem 4.1]. Mori forces a product relation among symmetries through unitary correlation matrices [17]. Slofstra embeds group relations in nonlocal games [25], and Paulsen and Rahaman obtain factorizable maps from bisynchronous game correlations [23].

1.3 Main results

Wang and Zhi have recently made this forced triviality quantitative [28], with explicit forms of the spectral correction of Alekseev, Liu and Thom and of Thom’s normalization theorem. Their reflection estimate holds uniformly in the matrix dimension. It concerns their auxiliary relation system, a finite set of group relations; Appendix A gives the form we use. Represent the root symbols by exact symmetries, and let every relation defect be below $2^{-\mathfrak{J}-7}$, where $\mathfrak{J} = 2^{2^{1122000}}$. Then two fixed conjugates of a root element lie within $5 \cdot 2^{-20}$ of each other [28, Theorem 5.2] (Theorem C.4). They use it to give an explicit polynomial counterexample to the algebraic form of Connes’ embedding conjecture. We use it, and their method, to give explicit channels. Their paper, and the preprints of Thom and of Alekseev, Liu and Thom that its proofs use, are not yet refereed. Section 1.4 lists which of our results depend on them. From Kun and Thom’s preprint we use only the groups.

Write $\mathcal{N}_d$ for the noisy operations on M_d , $\mathcal{K}_d$ for their closure, and Δ_J and $\Delta_\diamond$ for the normalized Choi and diamond distances to $\mathcal{K}_d$ (Section 2). An explicit positive lower bound on Δ_J or $\Delta_\diamond$ is called a *gap* below.

Theorem I (Section 4). *There is an explicit unital channel Φ_{12} on M_{4096} , factorizable through the group von Neumann algebra of the amalgamated double of a Kun–Thom pair over $\mathbb{F}_8$ used by Wang and Zhi, with Kraus entries in $\{0, 1, \pm 3/5, 4/5\}$ and Choi rank 949, such that*

$$\Delta_J(\Phi_{12}) \geq 2^{-(2\mathfrak{J}+50)}, \quad \Delta_\diamond(\Phi_{12}) \geq 2^{-(2\mathfrak{J}+50)}.$$

Theorem II (Theorem 3.4). *Take a finite relation system with relators of length at most six, and a tracial group model in which two generators p and q are distinct. Compile it into a rational channel Φ on M_d (Definition 3.1): one block for each prefix multiplication of each relator and for each generator inverse, and one singleton position for the identity and for each distinct value of a generator or inverse. Further singletons are allowed. Suppose that for some $0 < \rho_0 \leq 2$, in every matrix dimension, every unitary assignment with all relator defects $\|v(R) - I\|_2$ at most ρ_0 has $\|v_p - v_q\|_2 \leq \frac{1}{2}$. Then Φ is factorizable, as in [16, Proposition 3], and $\Delta_J(\Phi), \Delta_\diamond(\Phi) \geq (\rho_0/512)^2/d$.*

Theorem III (Section 7). *For the channels of [16], each reduced channel $\Phi \in \{\Phi_1, \Phi_2, \Phi_{15}\}$ satisfies $\Delta_J(\Phi), \Delta_\diamond(\Phi) \geq 2^{-(2\mathfrak{J}+41)}$. The fully referenced channel satisfies $\Delta_J(\Phi_{\text{full}}) \geq 2^{-(2\mathfrak{J}+36)}$ and $\Delta_\diamond(\Phi_{\text{full}}) \geq 2^{-(2\mathfrak{J}+21)}$, by the bound of [16, Theorem 6] (Lemma 7.3) with the tolerance of Theorem D.4.*

Theorem III gives explicit gaps for all four earlier channels. Φ_{12} acts on fewer qubits than any of them, but its gap is 2^9 times smaller than that of Φ_{15} . The two come from different relation systems, and neither bound is claimed to be optimal. The rigidity input of Theorem III is a new spectral certificate for the characteristic-five root groups of the 43-generator presentation (Appendix D). Theorem D.4 runs Wang and Zhi’s proof of their reflection estimate [28, Theorem 5.2] on this presentation, with their explicit forms of the spectral correction of Alekseev, Liu and Thom and of Thom’s normalization [28, Corollary 4.3 and Theorem 4.1]. The proof for the reduced channels uses a leakage theorem that needs no singletons for the generators (Theorem 7.1), a quantitative form of

[16, Lemma 2], and the Kraus-vector coherence test of [16, Theorem 5]. Table 1 places these results among the known separations.

Two further points. For Φ_{12} and for the earlier reduced channels, one fixed two-outcome measurement on the Choi state separates the channel from all of $\mathcal{K}_d$ (Corollary 5.1 and the remark after it). At errors below the gap, the gaps force linear purity consumption in repeated use. They also force linear memory when the exchanged qubits are capped (Section 6 for Φ_{12} ; the end of Section 7 for Φ_2 and Φ_{15}). The constants are astronomically small, inherited from the tolerance $2^{-\mathfrak{J}}$. They replace a compactness argument by an explicit bound; they do not describe an experiment.

1.4 What is new and what is not

The rational block of Figure 1 and the principle that a nearby finite-bath channel yields an approximate representation of the relations are from [16], in exact form [16, Lemma 2 and Proposition 3] and in quantitative form for fully referenced channels [16, Theorems 6 and I.1]. So are the coherence tests, between two singleton positions and between two Kraus vectors [16, Theorems 5 and I.1]. The singleton test is a quantitative form of the Haagerup–Musat criterion for Schur channels [8, Proposition 2.8] (Lemma 2.8 in the arXiv version). For a Schur channel, the unitary of an exact factorization is block diagonal, and its diagonal blocks are unitaries whose correlation matrix is the Schur matrix. With Wang and Zhi’s tolerance, the argument of Lemma 7.3, with the two signal singletons in place of the identity and the witness, would already give an explicit gap for a fully referenced channel of their relation system, on thirteen qubits (Remark 7.5). The compiler of Section 3 needs singletons only for the identity and the *letters*, the generators and their inverses, and recovers the relations from the multiplication blocks. This brings the channel to twelve qubits, with a diamond bound weaker by a factor of about 2^{13} . The new mathematics here is that compiler (Theorems 3.3 and 3.4), the leakage theorem without letter singletons (Theorem 7.1), and the characteristic-five root certificate (Appendix D); Theorem D.4 is Wang and Zhi’s argument run on that certificate. The explicit witness follows in a few lines from the proof of the compiler. The repeated-use bounds of Section 6 are applications of results from our companion papers.

Dependencies.

- No recent preprint is used in Theorems 3.3 and 3.4 (hence Theorem II), Lemma 3.2, Corollary 5.1, Theorem 7.1, Lemma 7.3, Proposition A.1, the certificate (9), Lemmas D.1 and D.2

	explicit object	separation	rigidity input
MIP* = RE [11]	none	existence	—
Liu [15]; Alekseev–Liu–Thom [1]	nonhyperlinear groups	witness forced trivial; no value	internality; Thom’s normalization
Mghirbi–Douglas [16]	rational channels, dimensions 31981 to 44086	positive, no value	Alekseev–Liu–Thom, Thom
Wang–Zhi [28]	degree-12 polynomial, 65 variables	$\text{tr } f \geq 3/4$ vs. -1	explicit forms of Alekseev–Liu–Thom and Thom
This paper	a 12-qubit channel, and the channels of [16]	explicit: $2^{-(2\mathfrak{J}+50)}$; $2^{-(2\mathfrak{J}+41)}$ and $2^{-(2\mathfrak{J}+36)}$	Wang–Zhi; new certificate

Table 1: Known failures of Connes’ embedding, from existence to explicit polynomials and channels. The gap $2^{-(2\mathfrak{J}+41)}$ holds for Φ_1 , Φ_2 and Φ_{15} ; the Choi-distance gap of Φ_{full} is listed last.

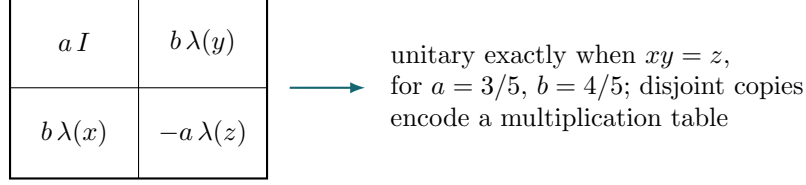


Figure 1: The rational block of [16]. It equals $\text{diag}(I, \lambda(x)) \left(\begin{pmatrix} a & b \\ b & -a \end{pmatrix} \otimes I \right) \text{diag}(I, \lambda(y))$ when $xy = z$.

and Proposition D.3.

- Proposition 4.1, Theorems 4.2, 7.2 and D.4, and Corollary 7.4 use Wang and Zhi’s theorems, and through them steps of Thom and of Alekseev, Liu and Thom. So do Theorems I and III, the explicit witness margins of Section 5, Corollary 6.1, the repeated-use statements at the end of Section 7 and Remark 7.5. These results depend on preprints that are not yet refereed.
- Lemma C.1 uses only Wang and Zhi’s parameter recipe and the arithmetic in their proof of Lemma 5.1.
- No result uses a theorem of Kun and Thom: we use their groups, and the facts we need about them are checked by exact computation.
- Corollary 6.1 and the end of Section 7 also rest on our companion reports [4, 5] and on [16, Proposition K.3], which are not refereed either.
- The bounds of Theorem III use the triangle table and Theorem D.4, not the group model of Section 7. The model gives factorizability; the scripts of Section 8 check its finite facts.

1.5 Organization

Section 2 fixes conventions. Section 3 proves the compiler theorems. Section 4 applies them to Wang and Zhi’s relation system, and Section 5 gives the explicit witness. Section 6 records the repeated-use consequences. Section 7 gives explicit gaps for the earlier channels. Section 8 describes the verification scripts, and Section 9 discusses open directions. The appendices give the relation system (A), the spectral certificates (B), the statements we use from Wang and Zhi (C), and the characteristic-five rigidity (D).

2 Setting

Let $M_d = M_d(\mathbb{C})$, let $\tau_m = \text{Tr}/m$ be the normalized trace on M_m , and write $\|X\|_2 = \tau_m(X^*X)^{1/2}$ for the normalized Hilbert–Schmidt norm and $\|X\|$ for the operator norm. A *noisy operation* on M_d is a channel of the form

$$\Psi(X) = (\text{id} \otimes \tau_m)[U(X \otimes I_m)U^*], \quad U \in U(dm), \quad (1)$$

for some m . Write $\mathcal{N}_d$ for this set and $\mathcal{K}_d$ for its closure. These are the channels with an exact factorization through $M_d \otimes M_m$ [9, Definition 3.1], the m -noisy channels of Musat’s survey, which uses “noisy operations” for a larger class [18]. The channels that factor through a finite-dimensional tracial von Neumann algebra are the finite convex mixtures of noisy operations, and they have the same closure [18, Proposition 3.8(ii)], so $\mathcal{K}_d$ does not depend on which finite baths are allowed.

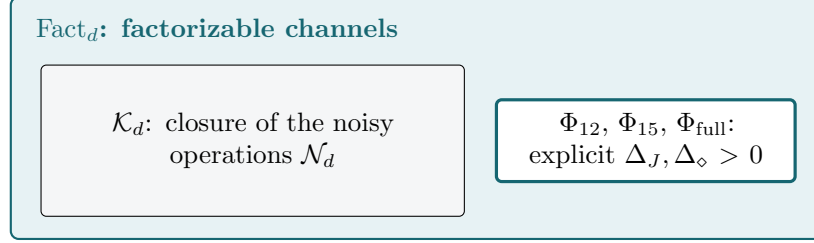


Figure 2: The classes of Section 2. The channels of this paper are factorizable and, using the preprints listed in Section 1.4, at an explicit positive distance from $\mathcal{K}_d$.

A channel is *factorizable* if it has the form (1) with (M_m, τ_m) replaced by a finite von Neumann algebra with a faithful normal tracial state; the class was introduced by Anantharaman-Delaroche [3, Definition 6.2] and characterized in this unitary form by Haagerup and Musat [8, Theorem 2.2]. Write Fact_d for this set. By a theorem of Haagerup and Musat, $\mathcal{K}_d$ consists of the channels that admit a factorization through a finite von Neumann algebra that embeds trace-preservingly into an ultrapower of the hyperfinite II_1 factor [9, Theorem 3.6]. Such a channel can have other ancillas that do not embed. So $\mathcal{N}_d \subseteq \mathcal{K}_d \subseteq \text{Fact}_d$ (Figure 2). Equality $\mathcal{K}_d = \text{Fact}_d$ for every d is equivalent to a positive answer to Connes' embedding problem [9, Theorem 3.7].

For $|\Omega_d\rangle = d^{-1/2} \sum_i |i, i\rangle$ let $J(\Phi) = (\Phi \otimes \text{id})(|\Omega_d\rangle\langle\Omega_d|)$ be the normalized Choi state. We use

$$d_J(\Phi, \Psi) = \frac{1}{2} \|J(\Phi) - J(\Psi)\|_1, \quad d_\diamond(\Phi, \Psi) = \frac{1}{2} \|\Phi - \Psi\|_\diamond,$$

and write $\Delta_J(\Phi)$ and $\Delta_\diamond(\Phi)$ for the distances from Φ to $\mathcal{K}_d$. Since a maximally entangled input is admissible in the diamond norm, $d_J \leq d_\diamond$. Haagerup and Musat measure distance in the cb norm of maps on M_d . By duality $\|\Phi - \Psi\|_\diamond = \|\Phi^* - \Psi^*\|_{\text{cb}}$, where Φ^* is the Hilbert–Schmidt adjoint, and the adjoint of a noisy operation is a noisy operation, with U^* in place of U . So $\Delta_\diamond(\Phi) = \frac{1}{2} \text{dist}_{\text{cb}}(\Phi^*, \mathcal{K}_d)$. The Choi distance does not change when both channels are replaced by their adjoints, so also $\text{dist}_{\text{cb}}(\Phi, \mathcal{K}_d) \geq 2\Delta_J(\Phi)$. For example, Haagerup and Musat's cb distance $4/27$ from W_3^- to the factorizable channels is a half-diamond distance of $2/27$. This holds because W_3^- is its own adjoint and adjoints of factorizable channels are factorizable [8, Remark 1.4(b)]. The closure $\mathcal{K}_d$ is the same in every norm. For an effect $0 \leq F \leq I$ and two channels, $d_J(\Phi, \Psi) \geq |\text{Tr } F(J(\Psi) - J(\Phi))|$.

3 From relations to rational channels

3.1 The compiled channel

A *relation system* is a finite set of positive generators and a finite set $\mathcal{R}$ of relators, words in the generators and their inverses; we call the generators and their inverses *letters*. A *tracial group model* is a homomorphism from the free group on the generators to a countable group Γ that kills every relator; we use it with the trace τ of the group von Neumann algebra and write λ for the left regular representation. The *labels* are the elements of Γ that occur below: the identity, the values of all letters, and the values of all relator prefixes. For example, take one generator s and the relator ss , with $\Gamma = \mathbb{Z}/2$. The relator has prefixes s and ss , with labels s and 1 ; since $s = s^{-1}$ in Γ , the letters s and s^{-1} have the same label. For a unitary assignment $s \mapsto v_s$ to the positive generators, the *defect* of a relator R is $\|v(R) - I\|_2$, where $v(R)$ evaluates R with $v_{s^{-1}} = v_s^*$.

Each multiplication $xy = z$ among labels is encoded by a *triangle*: the block of Figure 1, with $a = 3/5$, $b = 4/5$, on two system positions of its own. For a generator s , the triangle $(s, s^{-1}, 1)$ is its *inverse triangle*.

Definition 3.1 (Anchored-letter compiler). *The compiled channel has, on disjoint system positions,*

- (i) *a singleton position for the identity and one for every distinct letter label, and, optionally, singletons for further labels, each label having at most one singleton;*
- (ii) *a triangle for every prefix multiplication of every relator, read left to right, and the inverse triangle of every generator; equal triangles are merged.*

At a singleton for label g the Kraus operator A_g has entry 1. On a triangle (x, y, z) occupying positions $k, k+1$,

$$(A_1)_{kk} = a, \quad (A_y)_{k,k+1} = b, \quad (A_x)_{k+1,k} = b, \quad (A_z)_{k+1,k+1} = -a.$$

The channel is $\Phi(X) = \sum_g A_g X A_g^$, the sum over labels.*

In the example, the only triangle is $(s, s, 1)$, both a prefix triangle and the inverse triangle, and the compiled channel lives on four positions: singletons for 1 and s , and one block.

The block of Figure 1 is unitary because $a^2 + b^2 = 1$ and $xy = z$, so $U_\Gamma = \sum_g A_g \otimes \lambda(g)$ is unitary on $\mathbb{C}^d \otimes \ell^2(\Gamma)$. Distinct labels are distinct elements of Γ , so $\tau(\lambda(g)\lambda(h)^*) = \delta_{gh}$ and

$$(\text{id} \otimes \tau)[U_\Gamma(X \otimes 1)U_\Gamma^*] = \sum_g A_g X A_g^* = \Phi(X).$$

Thus Φ is unital, trace preserving and factorizable, with Choi rank equal to the number of labels. This is the block factorization of [16, Proposition 3], an instance of the coefficient criterion of Haagerup and Musat [8, Theorem 2.2], written in our convention $U(X \otimes 1)U^*$.

3.2 Extraction from small support leakage

Let Π be the support projection of $J(\Phi)$. For a noisy operation Ψ with unitary U as in (1), write $U = \sum_{ij} |i\rangle\langle j| \otimes U_{ij}$ with bath blocks $U_{ij} \in M_m$. From the definitions,

$$J(\Psi) = \frac{1}{d} \sum_{i,j,k,l} \tau_m(U_{ij}U_{kl}^*) |i\rangle\langle k| \otimes |j\rangle\langle l|,$$

the Gram matrix of the d^2 bath blocks in $L^2(M_m, \tau_m)$, divided by d . Identify $|i\rangle \otimes |j\rangle$ with the matrix unit E_{ij} . The Kraus matrices have pairwise disjoint supports, so they are orthogonal, and Π is the projection onto their span. Projecting U , as an element of $M_d \otimes L^2(M_m, \tau_m)$, onto $\text{span}\{A_g\} \otimes L^2$ gives

$$U = W + R, \quad W = \sum_g A_g \otimes B_g, \quad B_g = \frac{\sum_{ij} (A_g)_{ij} U_{ij}}{\sum_{ij} (A_g)_{ij}^2}, \quad (2)$$

and since the supports are disjoint, $W_{ij} = (A_g)_{ij} B_g$ for the one label g whose Kraus matrix has entry (i, j) , and $W_{ij} = 0$ if there is none. For an exact factorization $R = 0$ and the B_g are orthonormal, as in the proof of [8, Theorem 2.2]; the leakage measures how far a noisy operation is from this. Define the *support leakage*, as in [4, Definition 2.2],

$$\nu = \text{Tr}[(I - \Pi)J(\Psi)], \quad r = \sqrt{d\nu}; \quad \text{then} \quad \sum_{ij} \|R_{ij}\|_2^2 = d\nu = r^2.$$

Lemma 3.2 (Anchors; compare [16, Appendix I.1]). *Let s be a singleton position for the label g , and let V_s be a unitary polar part of the diagonal block U_{ss} . Then $\tau(U_{ss}U_{ss}^*) \geq 1 - r^2$, $\|U_{ss} - V_s\|_2 \leq r$ and $\|B_g - V_s\|_2 \leq 2r$.*

Proof. In row and column s the projection W has only the entry $W_{ss} = B_g$, so the other entries of that column belong to R . Column unitarity gives $U_{ss}^* U_{ss} = I - P$ with $P = \sum_{k \neq s} U_{ks}^* U_{ks}$, so $0 \leq P \leq I$ and $\|P\|_2^2 \leq \tau(P) \leq r^2$. Hence $\tau(U_{ss} U_{ss}^*) = \tau(U_{ss}^* U_{ss}) \geq 1 - r^2$, and since $1 - t \leq 1 - t^2$ on $[0, 1]$, $\|U_{ss} - V_s\|_2 = \|I - |U_{ss}\|_2 \leq \|P\|_2 \leq r$. Finally $\|B_g - U_{ss}\|_2 = \|R_{ss}\|_2 \leq r$. $\square$

Replacing U by $(I \otimes V_1^*)U$, where V_1 is the polar part at the identity singleton, leaves Ψ unchanged. After this normalization $V_1 = I$ and $\|B_1 - I\|_2 \leq 2r$. For each letter s put $v_s = V_s$, the polar part at the singleton of its label; for an inverse letter this is a unitary $v_{s^{-1}}$ of its own. For a word w let $\hat{v}(w)$ be the product of these letter unitaries.

Theorem 3.3 (Extraction). *Let the relators have length at most six. For every noisy operation Ψ and every unitary U implementing it as in (1), normalized as above, the assignment $s \mapsto v_s$ to the positive generators has every relator defect at most $385r$.*

Proof. Fix a relator. Let w be one of its prefixes, with label x , and let y be the next letter, so that wy has label z and (x, y, z) is a triangle on positions $k, k+1$. Suppose $\|B_x - \hat{v}(w)\|_2 \leq cr$. (This is a statement about this relator; the coefficient B_x is shared by every occurrence of the label x .) The four blocks are

$$U_{kk} = aB_1 + R_{kk}, \quad U_{k,k+1} = bB_y + R_{k,k+1}, \quad U_{k+1,k} = bB_x + R_{k+1,k}, \quad U_{k+1,k+1} = -aB_z + R_{k+1,k+1}.$$

Columns k and $k+1$ of U are orthogonal:

$$U_{kk}^* U_{k,k+1} + U_{k+1,k}^* U_{k+1,k+1} + O = 0, \quad O = \sum_{i \neq k, k+1} U_{ik}^* U_{i,k+1}.$$

The entries U_{ik} with $i \neq k, k+1$ form a column operator of norm at most one, whose normalized 2-norm is at most r because these entries belong to R ; so $\|O\|_2 \leq r$. This is the leakage form of the row estimate of [16, Appendix I.2]. Now

- $\|U_{k+1,k} - b\hat{v}(w)\|_2 \leq (bc+1)r$ and $\|U_{k+1,k+1}\| \leq 1$, so $\|U_{k+1,k}^* U_{k+1,k+1} - b\hat{v}(w)^* U_{k+1,k+1}\|_2 \leq (bc+1)r$;
- $\|U_{kk} - aI\|_2 \leq (2a+1)r$ and $\|U_{k,k+1} - bv_y\|_2 \leq (2b+1)r$ by Lemma 3.2, so

$$\|U_{kk}^* U_{k,k+1} - abv_y\|_2 \leq \|U_{kk} - aI\|_2 + a\|U_{k,k+1} - bv_y\|_2 \leq (2a+1 + a(2b+1))r = \frac{94}{25}r.$$

Hence $\|b\hat{v}(w)^* U_{k+1,k+1} + abv_y\|_2 \leq (bc+1 + 1 + \frac{94}{25})r = (bc + \frac{144}{25})r$. Dividing by b and multiplying by $\hat{v}(w)$ gives $\|U_{k+1,k+1} + a\hat{v}(w)v_y\|_2 \leq (c + \frac{36}{5})r$, and comparing $U_{k+1,k+1}$ with $-aB_z$ gives

$$\|B_z - \hat{v}(wy)\|_2 \leq \frac{c + \frac{36}{5} + 1}{a} r = \left(\frac{5}{3}c + \frac{41}{3}\right)r.$$

The first prefix is a letter, with $c_1 = 2$ by Lemma 3.2, so along a relator of length at most six the constants are at most

$$c_1 = 2, \quad c_2 = 17, \quad c_3 = 42, \quad c_4 = \frac{251}{3}, \quad c_5 = \frac{1378}{9}, \quad c_6 = \frac{7259}{27}.$$

The full relator has label 1, and $\|B_1 - I\|_2 \leq 2r$, so $\|\hat{v}(R) - I\|_2 \leq (c_6 + 2)r = \frac{7313}{27}r$.

It remains to pass from $\hat{v}(R)$ to $v(R)$, replacing each $v_{s^{-1}}$ by v_s^* . The inverse triangle $(s, s^{-1}, 1)$ is a prefix triangle of length two, so $\|v_s v_{s^{-1}} - I\|_2 \leq (c_2 + 2)r = 19r$ and $\|v_{s^{-1}} - v_s^*\|_2 \leq 19r$. A relator has at most six inverse letters, and each replacement costs at most $19r$ in $\|\cdot\|_2$ by unitary invariance. The total is at most $(\frac{7313}{27} + 114)r = \frac{10391}{27}r < 385r$. $\square$

For relators of length ℓ the same recursion gives $c_\ell = \frac{45}{2}(\frac{5}{3})^{\ell-1} - \frac{41}{2}$; we only use $\ell \leq 6$. The letter singletons are a convenience: the method of Theorem 7.1 recovers letter unitaries from the triangles alone, with larger constants.

3.3 The gap

Theorem 3.4 (Compiler gap). *Let Φ be compiled from a relation system with relators of length at most six and a tracial group model. Let p, q be generators with $p \neq q$ in the model, and suppose that for some $0 < \rho_0 \leq 2$, in every matrix dimension,*

$$\text{every unitary assignment with all relator defects at most } \rho_0 \text{ has } \|v_p - v_q\|_2 \leq \frac{1}{2}. \quad (3)$$

Then $\Phi \in \text{Fact}_d$ and

$$\Delta_J(\Phi) \geq g, \quad \Delta_\diamond(\Phi) \geq g, \quad g = \left(\frac{\rho_0}{512}\right)^2 \frac{1}{d}.$$

Proof. Factorizability was shown above. Let $\Psi \in \mathcal{N}_d$ have leakage ν . Since $\text{Tr}[(I - \Pi)J(\Phi)] = 0$, the effect $I - \Pi$ gives $d_J(\Phi, \Psi) \geq \nu$. So assume $\nu < g$, that is $r < \rho_0/512 \leq 1/256$. Theorem 3.3 gives relator defects below $385r < \rho_0$, so $\|v_p - v_q\|_2 \leq 1/2$ by (3).

Let s_p, s_q be the singleton positions of p, q and $X_p = U_{s_p s_p}$, $X_q = U_{s_q s_q}$. By Lemma 3.2, $\|X_p - v_p\|_2, \|X_q - v_q\|_2 \leq r$, and $\|X_p\|, \|X_q\| \leq 1$, so

$$\Re \tau(X_p X_q^*) \geq \Re \tau(v_p v_q^*) - 2r = 1 - \frac{1}{2} \|v_p - v_q\|_2^2 - 2r \geq \frac{7}{8} - 2r.$$

Let F_{coh} be the projection onto $(|s_p, s_p\rangle + |s_q, s_q\rangle)/\sqrt{2}$. Since $\langle s, s | J(\Psi) | s', s' \rangle = d^{-1} \tau(U_{ss} U_{s's'}^*)$,

$$\text{Tr } F_{\text{coh}} J(\Psi) = \frac{1}{2d} \left(\tau(X_p X_p^*) + \tau(X_q X_q^*) + 2 \Re \tau(X_p X_q^*) \right) \geq \frac{1}{d} \left(1 - r^2 + \frac{7}{8} - 2r \right).$$

For the target, $\Phi(|s_p\rangle\langle s_q|) = 0$ because the labels differ, so $\text{Tr } F_{\text{coh}} J(\Phi) = 1/d$. Hence $d_J(\Phi, \Psi) \geq (\frac{7}{8} - r^2 - 2r)/d > \frac{3}{4d} > g$. Taking the infimum over $\mathcal{N}_d$ and passing to the closure gives $\Delta_J(\Phi) \geq g$, and $d_J \leq d_\diamond$ gives the diamond bound. $\square$

4 The twelve-qubit channel

We apply Theorem 3.4 to the relation system of Wang and Zhi, described in Appendix A. It has 81 positive generators: 72 involutive root symbols, seven action symbols and two signals p, q , which are involutions in the model. Its 917 relators have length at most six. The tracial group model is the amalgamated double $\Lambda = G *_H G$ of a Kun–Thom pair over $\mathbb{F}_8$ [14, Theorem E], which Wang and Zhi use. In it there are 949 labels, of which 88 are letter labels, and 1866 triangles (Proposition A.1). The signals are distinct; in the shipped data they are the labels with indices 87 and 88.

The triangles occupy 3732 positions. The remaining 364 positions of $\mathbb{C}^{4096}$ are singletons for 364 distinct labels: the identity, the 88 letter labels, and 275 further labels. This is the channel Φ_{12} . Its Kraus operators have 7828 nonzero entries, and its Choi rank is 949. The extra singletons serve only Section 6.

Why p and q must be close in matrices. In Λ the signals $p = U_{12}^{-1} h U_{12}$ and $q = t^{-1} h t$, where $h = z_{2,0}$, lie outside H in different factors of the amalgam, so they differ [28, (2.9) and Lemma 2.2]. For exact solutions in matrices the reason they agree is simple: a unitary conjugation that maps a matrix subalgebra into itself maps it onto itself, so two conjugations that agree on the algebra generated by the roots also agree in reverse, which forces $U_{12}^{-1} h U_{12} = t^{-1} h t$ [28, Lemma 2.1]; compare Thom’s observation that a compressor normalizes every finite image of the subgroup [27, Remark 4.4] and [16, Lemma 4]. For approximate solutions Wang and Zhi make this argument quantitative, with a spectral-gap argument in place of the dimension count. Spectral gap gives a

subalgebra $\mathcal{A}$ close to the relative commutant of the roots, as in the internality theorem of Alekseev, Liu and Thom (Theorem C.2). The compression relations give a near-inclusion $U^*\mathcal{A}U \subseteq \mathcal{A}$, which Wang and Zhi's explicit form of Thom's normalization theorem turns around (Theorem C.3). The common compression targets then force the two conjugates together, uniformly in the dimension, as in Thom's proof of his Theorem 1.3 (Theorem C.4).

Let $\delta = 2^{-\mathfrak{J}}$ with $\mathfrak{J} = 2^{2^{1122000}}$ as in [28, (2.20)].

Proposition 4.1 (Rigidity tolerance). *Put $\rho_0 = \delta/2^{10}$. Every unitary assignment to the 81 generators with all 917 relator defects at most ρ_0 has $\|v_p - v_q\|_2 < 2^{-17}$, in every matrix dimension.*

Proof. Wang and Zhi's estimate needs exact symmetries for the roots. For a root generator s , let $\tilde{v}_s$ be the symmetry obtained from v_s by spectral calculus, rounding each eigenvalue $e^{i\theta}$ to the nearer of ± 1 , and to $+1$ when $\Re e^{i\theta} = 0$ (compare the rounding in [28, Lemma 5.4]). Since $|e^{i\theta} \mp 1| \leq |e^{2i\theta} - 1|$ for the nearer sign, $\|v_s - \tilde{v}_s\|_2 \leq \|v_s^2 - I\|_2 \leq \rho_0$. Each balanced equation of Wang and Zhi's system has at most six letters, so after rounding its defect is at most $\rho_0 + 6\rho_0 = 7\rho_0 < \delta/128$. The 843 balanced equations of our system are those of Wang and Zhi's auxiliary system [28, Section 5.1], with $p_{ab,r}$ and $p_{ba,r}$ identified. Giving both ordered symbols the rounded value of the identified one, as Wang and Zhi do in the proof of their Lemma 5.4, yields an assignment that satisfies the hypotheses of their Theorem 5.2 (Theorem C.4):

$$\|U_{12}^* \tilde{h} U_{12} - t^* \tilde{h} t\|_2 \leq 5 \cdot 2^{-20}, \quad \tilde{h} = \tilde{v}_{z_{2,0}}.$$

The naming relators $p = U_{12}^{-1} h U_{12}$ and $q = t^{-1} h t$ have defect at most ρ_0 , and $\|v_h - \tilde{h}\|_2 \leq \rho_0$, so $\|v_p - v_q\|_2 \leq 5 \cdot 2^{-20} + 4\rho_0 < 2^{-17}$. $\square$

Wang and Zhi's proof of Theorem 5.2 uses an integer certificate whose data are available from them on request [28, proof of Lemma 3.3]. The independently computed certificate of Appendix B can be used in its place, since Lemma C.1 covers its constant.

Theorem 4.2 (Theorem I). $\Phi_{12} \in \text{Fact}_{4096}$ and $\Delta_J(\Phi_{12}), \Delta_\diamond(\Phi_{12}) \geq g_* = 2^{-(2\mathfrak{J}+50)}$.

Proof. Apply Theorem 3.4 with $\rho_0 = 2^{-(\mathfrak{J}+10)}$ from Proposition 4.1: $g = (2^{-(\mathfrak{J}+19)})^2/2^{12} = 2^{-(2\mathfrak{J}+50)}$. $\square$

We obtain the diamond bound only from $d_J \leq d_\diamond$, so it carries the factor $1/d$. By [9, Theorem 3.6], Theorem 4.2 also shows that no ancilla of Φ_{12} embeds trace-preservingly into an ultrapower of the hyperfinite II_1 factor. The converse fails in general: a factorization through a non-embeddable algebra, such as the group von Neumann algebra of Λ , does not by itself put a channel outside $\mathcal{K}_d$, because ancillas are far from unique [18, Example 3.5].

5 An explicit witness

Let Φ satisfy the hypotheses of Theorem 3.4, with $g = (\rho_0/512)^2/d$. Consider the two effects

$$F_{\text{leak}} = I - \Pi, \quad F_{\text{coh}} = \frac{1}{2}(|s_p, s_p\rangle + |s_q, s_q\rangle)(\langle s_p, s_p| + \langle s_q, s_q|),$$

with target values $\text{Tr } F_{\text{leak}} J(\Phi) = 0$ and $\text{Tr } F_{\text{coh}} J(\Phi) = 1/d$. The proof of Theorem 3.4 shows that for every $\Psi \in \mathcal{N}_d$ either $\text{Tr } F_{\text{leak}} J(\Psi) \geq g$ or $\text{Tr } F_{\text{coh}} J(\Psi) > (1 + \frac{3}{4})/d$. One effect combines both tests.

Corollary 5.1 (Explicit witness). *Put $\alpha = dg/2$ and $F_* = (F_{\text{leak}} + \alpha F_{\text{coh}})/(1 + \alpha)$. Then $0 \leq F_* \leq I$, and for every $\Psi \in \mathcal{K}_d$,*

$$\text{Tr } F_* J(\Psi) - \text{Tr } F_* J(\Phi) \geq \frac{3g}{8(1 + \alpha)}.$$

Proof. For $\Psi \in \mathcal{N}_d$ put $\ell = \text{Tr } F_{\text{leak}} J(\Psi)$ and $\kappa = \text{Tr } F_{\text{coh}} J(\Psi) - 1/d \geq -1/d$. If $\ell \geq g$ then $\ell + \alpha\kappa \geq g - \alpha/d = g/2$. If $\ell < g$ then $\kappa > 3/(4d)$ and $\ell + \alpha\kappa > 3g/8$. Divide by $1 + \alpha$. The bound passes to the closure by continuity. $\square$

For Φ_{12} the margin is about $2^{-(2\mathfrak{J}+51.4)}$. Because α is tiny, F_* is essentially the leakage projection, a projection of rank $4096^2 - 949$ on the twenty-four qubits of system and reference. A separating effect exists for any channel with $\Delta_J > 0$, since $\mathcal{K}_d$ is closed and convex; Corollary 5.1 says only that this one is explicit. The same construction works for the reduced channels of Section 7, with the Kraus-vector effect F of Theorem 7.2 in place of F_{coh} , but the weight must change: the target value is now $\text{Tr } FJ(\Phi) = (s_1 + s_j)/(2d)$ rather than $1/d$. With $g = \delta_5^2/(2^{26}d)$ as in Theorem 7.2, put $\alpha = dg/(s_1 + s_j)$, $F_* = (F_{\text{leak}} + \alpha F)/(1 + \alpha)$ and $\kappa = \text{Tr } F(J(\Psi) - J(\Phi))$. If $\ell \geq g$ then $\ell + \alpha\kappa \geq g/2$. If $\ell < g$, the proof of Theorem 7.2 gives $\kappa > 1/(2d)$, hence $\ell + \alpha\kappa > g/(2(s_1 + s_j))$. The separation margin is therefore at least $g/(2(s_1 + s_j)(1 + \alpha))$. For all three channels $s_1 + s_j < 2^{13}$ (it is $164013/25$ for Φ_{15}), so the margin exceeds $g/2^{15} \geq 2^{-(2\mathfrak{J}+56)}$.

6 Repeated use

We recall the resource model of our companion paper [4, Definition 2.3]. A *causal service* of a channel Φ for n rounds is a device with

- a memory of Q qubits, starting in a state β_0 with *purity deficit* $P = Q - S(\beta_0)$ (qubits minus entropy, in bits);
- in each round, one unitary on the incoming system and the memory, after which the output is released before the next input arrives;
- between rounds, an exchange in which memory qubits are exported and each is replaced by a qubit of a fixed imported register, independent of everything else; E counts the exported qubits (equivalently, the imported ones), and C is the total purity deficit of the imports.

The companion paper writes J for E . The error ε is the largest half trace distance between the real and the ideal final states, over all adaptive observers who keep quantum references. A state is *flat* if its nonzero eigenvalues are equal.

Two results from earlier work apply directly. First, the purity floor [4, Theorem 3.2(b)], proved from local tracialization [4, Lemma 5.4] and an entropy ledger [4, Lemma 3.5]: for every exchange E and memory size Q ,

$$P + C \geq \frac{n(\Delta_J(\Phi) - \varepsilon)_+^2}{2 \ln 2}. \quad (4)$$

Second, the flat-probe budget [5, Proposition 13.1], also [16, Proposition K.3]: if a pure input-reference state has flat output of rank k , every causal service with $\varepsilon \leq 1/16$ and $E \leq \frac{n}{2} \log_2 k$ satisfies $P + C < 3Q + 6$.

For Φ_{12} the 364 singletons carry distinct labels, so on their span Φ_{12} is complete dephasing. The input $364^{-1/2} \sum_s |s, s\rangle$, summed over singleton positions, therefore has flat output of rank 364.

Corollary 6.1. *Every causal service of Φ_{12} satisfies $P + C \geq n(g_* - \varepsilon)_+^2 / (2 \ln 2)$. If moreover $\varepsilon \leq 1/16$ and $E \leq \frac{n}{2} \log_2 364$, then*

$$Q > \frac{n(g_* - \varepsilon)_+^2}{6 \ln 2} - 2.$$

Proof. The first bound is (4) with Theorem 4.2. Combined with $P + C < 3Q + 6$ it gives the second. $\square$

The exchange cap matters: above it, at exchange rates near the threshold of [4, Theorem 3.4], imported purity can replace memory, with sublinear memory at vanishing error [4, Corollary 3.3(c)]. At the larger cap $E \leq nH(\Phi)/2$, with H the maximal complementary entropy, linear memory follows for every channel outside $\mathcal{K}_d$ [4, Corollary 3.3(b)], for errors below about $\Delta_J^2 / (4 \ln 2 \cdot \log_2 d)$. At the scale of g_* these bounds are formal: they need errors below g_* , and the memory bound becomes positive only once n exceeds about $2^{4\mathfrak{J}+103}$.

7 Explicit gaps for the earlier channels

The channels of [16] come from a finite presentation modelled on the Kun–Thom pair over $\mathbb{F}_5$ [14, Theorem E] and Thom’s witness commutator [27, Section 5.2]. Let $\Sigma = (0, 1, 2, 3, -1, -2, -3)$. The generators are root generators f_r , for $f \in \{a, b, c\}$ and $r \in \Sigma$, and six action generators t_{ij} , $1 \leq i \neq j \leq 3$. The relations are fifth powers and commutation within each root family; the class-two relations

$$[[f_r, g_s], f_u] = [[f_r, g_s], g_u] = 1 \quad ((f, g) = (a, b), (b, c), (c, a); r, s, u \in \Sigma), \quad (5)$$

with $[x, y] = xyx^{-1}y^{-1}$; the Steinberg relations $[t_{ij}, t_{kl}] = 1$ for $i \neq l, j \neq k$, and $[t_{ij}, t_{jk}] = t_{ik}$ for distinct i, j, k ; for $\epsilon = \pm 1$ the compression relations

$$t_{ij}^\epsilon f_r t_{ij}^{-\epsilon} = \begin{cases} \beta_f(\epsilon i, j), & r = j, \\ \beta_f(-\epsilon i, -j), & r = -j, \\ f_r, & r = 0 \text{ or } |r| \neq j, \end{cases} \quad \beta_f(r, s) = [[f_r, (f^+)_0], [(f^-)_s, f_0]], \quad (6)$$

where f^+ and f^- are the cyclic successor and predecessor of f ; and $f_{-i} = \zeta_i f_i \zeta_i^{-1}$ with $\zeta_i = (t_{ik} t_{ki}^{-1} t_{ik})^2$ and $k = i + 1$ cyclically. Trivial and duplicate commuting relations are omitted.

Take two copies of this group, identify their twelve positive root generators $f_r, r \in \{0, 1, 2, 3\}$, and adjoin the witness $j = [h, t_1 t_0^{-1}]$ as a named generator, where $h = a_2$ and t_0, t_1 are the two copies of t_{12} . This gives 43 generators and 4417 freely reduced relators, of length at most 19. The relators are shipped as `earlier/presentation.json`; Appendices C and D of [16] give the enumeration conventions.

A tracial group model is the following [16, Section 2.2]. Let $R_+ = \mathbb{F}_5[x_1, x_2, x_3]$, $R = \mathbb{F}_5[x_1^{\pm 1}, x_2^{\pm 1}, x_3^{\pm 1}]$ and $G_0 = \text{SL}_3(R) \rtimes \text{SL}_3(\mathbb{Z})$, the second factor acting on exponent columns by monomial substitutions. Here SL_3 replaces the groups EL_3 of the Kun–Thom pair, as in [16]; the generators map into EL_3 , and the larger groups avoid identifying the image of the root subgroup with $\text{EL}_3(R_+)$. The product is $(M, A)(N, B) = (M\sigma_A(N), AB)$, where σ_A acts on Laurent monomials by $x^v \mapsto x^{Av}$. Write $e_{kl}(z) = I + zE_{kl}$. Map a_r, b_r, c_r to $e_{12}(z_r), e_{23}(z_r), e_{31}(z_r)$, where $z_0 = 1$ and $z_{\pm i} = x_i^{\pm 1}$, and map t_{ij} to the action matrix $I + E_{ij}$. All relations hold [16, Lemma C.1]; for instance the image of ζ_i is -1 on coordinates i and k , so it inverts x_i . The shared positive roots map into $H_0 = \text{SL}_3(R_+) \times \{I\}$ in both copies, so the double maps to $D_0 = G_0 *_{H_0} G_0$, and j

maps to the word $(ht)_1(t^{-1}h^{-1}t)_0(t^{-1})_1$. Its outer syllables have action $I \pm E_{12} \neq I$ and its middle syllable is $e_{12}(-x_1^{-1}x_2)$, so all three lie outside H_0 , the word is reduced, and $j \neq 1$ in the image Γ . This is Thom's normal-form argument [27, Section 5.2]. The channels factor through the group von Neumann algebra of Γ [16, Proposition F.1], and their labels are distinct elements of Γ [16, Appendix D]. The bounds below use only the triangle table and Theorem D.4, not this model.

For a letter y write $y^\vee$ for the label of the inverse letter. The original triangle table has 16428 triangles: the prefix triangles of all relators (no proper prefix of a relator is the identity here) and the inverse triangles $(s, s^{-1}, 1)$ of the 43 positive generators. Its fully referenced channel Φ_{full} , on M_{44086} , has a singleton for each of its 11230 labels. The channel Φ_{15} on M_{215} keeps 15990 of the triangles, deleting

- (a) 408 triangles $(z, y^\vee, x)$ for which (x, y, z) is retained and the inverse triangle of the generator underlying y is retained; that triangle reads $(y, y^\vee, 1)$ or $(y^\vee, y, 1)$, according to whether y is a positive or a negative letter;
- (b) 30 triangles $(y, x, 1)$ whose reverse $(x, y, 1)$ is an inverse triangle.

It has singletons only for the identity (787 positions) and the witness label (1 position). The reduced channel Φ_2 , on M_{31982} , is Φ_{15} without its 786 padding positions, and Φ_1 , on M_{31981} , also drops the witness singleton. Every label occurs in a retained triangle: the labels of a deleted triangle occur in the retained triangles that certify its deletion. All 43 inverse triangles are retained. The decomposition (2) and Lemma 3.2 use only that the Kraus matrices have disjoint supports and that a singleton is alone in its row and column, so they apply to these channels.

Theorem 7.1 (Leakage extraction without letter singletons). *Let $\Phi \in \{\Phi_1, \Phi_2, \Phi_{15}\}$ on M_d , let $\Psi \in \mathcal{N}_d$ have leakage ν with respect to Φ , with $r \leq 1$, and let U implement Ψ as in (1), normalized so that one fixed identity singleton has polar part I . There are unitaries v_g for all labels, with $v_1 = I$, $\|B_1 - I\|_2 \leq 2r$ and $\|B_g - v_g\|_2 \leq 34r$ for every label g , such that every relator of the 43-generator presentation has defect below $2^{13}r$ under the assignment $s \mapsto v_s$ to the positive generators.*

Proof. Letters from triangles. Take a triangle (x, y, z) on positions $k, k+1$ with blocks as in the proof of Theorem 3.3. Column k gives $U_{kk}^* U_{kk} + U_{k+1,k}^* U_{k+1,k} = I - O_1$, where $O_1 = C^*C$ for the column operator C of the entries outside rows $k, k+1$; so $\|O_1\|_2 \leq \|C\| \|C\|_2 \leq r$. Since $\|U_{kk} - aI\|_2 \leq \frac{11}{5}r$ and $\|U_{kk}\| \leq 1$,

$$\|U_{kk}^* U_{kk} - a^2 I\|_2 \leq (\|U_{kk}\| + a) \|U_{kk} - aI\|_2 \leq \frac{8}{5} \cdot \frac{11}{5} r = \frac{88}{25} r.$$

So $Y = U_{k+1,k}/b$ satisfies $\|Y^*Y - I\|_2 \leq (\frac{88}{25} + 1)r/b^2 = \frac{113}{16}r$. For its polar part V , since $|t-1| \leq |t^2-1|$ for $t \geq 0$, $\|Y - V\|_2 = \||Y| - I\|_2 \leq \|Y^*Y - I\|_2 \leq \frac{113}{16}r$; and $\|B_x - Y\|_2 \leq r/b = \frac{5}{4}r$. Row k treats $U_{k,k+1}$ in the same way. Hence every label occurring as x or y in a retained triangle has a unitary v_g with $\|B_g - v_g\|_2 \leq \frac{133}{16}r$. For the identity take $v_1 = I$, which satisfies $\|B_1 - v_1\|_2 \leq 2r$.

Labels occurring only as z . Repeating the computation in the proof of Theorem 3.3 with $\|B_x - v_x\|_2, \|B_y - v_y\|_2 \leq cr$ gives

$$\|B_z - v_x v_y\|_2 \leq \frac{1}{a} \left(\frac{3 + 3a + bc(1+a)}{b} + 1 \right) r. \quad (7)$$

With $c = \frac{133}{16}$ this is $\frac{203}{6}r < 34r$. Put $v_z = v_x v_y$.

Retained triangles. Now $\|B_g - v_g\|_2 \leq \frac{203}{6}r$ for every label, and (7) with $c = \frac{203}{6}$, plus $\|B_z - v_z\|_2$, gives $\|v_z - v_x v_y\|_2 \leq \frac{2443}{18}r < 136r$.

Deleted triangles. For (a), with (x, y, z) retained,

$$\|v_z v_{y^\vee} - v_x\|_2 \leq \|(v_z - v_x v_y) v_{y^\vee}\|_2 + \|v_y v_{y^\vee} - I\|_2 < 136r + 136r = 272r,$$

because $\|v_y v_{y^\vee} - I\|_2 = \|v_{y^\vee} v_y - I\|_2$ for unitaries, so either orientation of the retained inverse triangle bounds it. For (b), $\|v_y v_x - I\|_2 = \|v_x^* (v_x v_y - I) v_x\|_2 < 136r$.

Relators. A relator of length at most 19 has at most 18 prefix merges. Each is a retained or deleted triangle, except that a merge from an identity prefix would need no triangle and cost nothing, since $v_1 = I$; none occurs here. The full prefix has label 1. Telescoping gives letter-evaluation defect below $18 \cdot 272r$. Replacing each of at most 19 inverse letters by the adjoint of its positive generator costs below $136r$ each, by the inverse triangles. The total is below $18 \cdot 272r + 19 \cdot 136r = 7480r < 2^{13}r$. $\square$

Let $\delta_5 = 2^{-3}$ be the tolerance of Theorem D.4 for the 43-generator presentation.

Theorem 7.2 (Theorem III, reduced channels). *Let $\Phi \in \{\Phi_1, \Phi_2, \Phi_{15}\}$ on M_d , with $d = 31981$, 31982 or 2^{15} . Then $\Delta_J(\Phi), \Delta_\diamond(\Phi) \geq \delta_5^2 / (2^{26}d) \geq 2^{-(23+41)}$.*

Proof. Put $g = \delta_5^2 / (2^{26}d)$ and let $\Psi \in \mathcal{N}_d$ have leakage ν . If $\nu \geq g$ the support effect gives $d_J \geq g$. Otherwise $r < 2^{-13}\delta_5$, so Theorem 7.1 gives relator defects below δ_5 , and Theorem D.4 gives $\|v_j - I\|_2 < 2^{-17}$ for the witness generator j .

We use a coherence test on Kraus vectors, as in [16, Theorem 5]. Let $s_g = \|A_g\|_{\text{HS}}^2$, so $\sum_g s_g = \text{Tr} \sum_g A_g^* A_g = d$, and let $|a_g\rangle = \text{vec}(A_g) / \sqrt{s_g}$. These vectors are orthogonal, and from (2),

$$\langle a_g | J(\Psi) | a_h \rangle = \frac{\sqrt{s_g s_h}}{d} \tau(B_g B_h^*), \quad \langle a_g | J(\Phi) | a_h \rangle = \frac{s_g}{d} \delta_{gh}.$$

Let F be the projection onto $(|a_1\rangle + |a_j\rangle) / \sqrt{2}$. Then

$$\text{Tr} F(J(\Psi) - J(\Phi)) = \frac{1}{2d} \left(s_1 (\|B_1\|_2^2 - 1) + s_j (\|B_j\|_2^2 - 1) + 2\sqrt{s_1 s_j} \Re \tau(B_1 B_j^*) \right).$$

By Theorem 7.1, $\|B_g\|_2 \geq 1 - 34r$ for $g = 1, j$, so the first two terms are at least $-68r(s_1 + s_j) / (2d) \geq -34r$. Also $\|B_j\|_2 \leq 1 + 34r \leq \frac{3}{2}$, as $r < 2^{-13}\delta_5$, so

$$\Re \tau(B_1 B_j^*) \geq \Re \tau(v_j^*) - \|B_1 - I\|_2 \|B_j\|_2 - \|B_j - v_j\|_2 \geq 1 - 2^{-35} - 37r.$$

The identity singleton gives $s_1 \geq 1$, and every nonzero Kraus entry has modulus at least $3/5$, so $\sqrt{s_1 s_j} \geq 3/5$. Hence $\text{Tr} F(J(\Psi) - J(\Phi)) \geq \frac{3}{5d} (1 - 2^{-35} - 37r) - 34r > \frac{1}{2d} > g$. Take the infimum, the closure, and $d_J \leq d_\diamond$. Finally $d \leq 2^{15}$ gives $g \geq 2^{-(23+41)}$. $\square$

For the fully referenced channel we use the following bound of [16]. Call $e_0 \in (0, 1]$ a *relator tolerance* for a presentation with witness j if, in every matrix dimension, every unitary assignment to the generators with all relator defects at most e_0 has $\|v_j - I\|_2 \leq \frac{1}{2}$.

Lemma 7.3 (Fully referenced channels [16, Theorem 6]). *Let Φ on M_d be built by the block rule of Definition 3.1 from a presentation with a distinguished generator j , the witness, and a tracial group model in which $j \neq 1$ (the model gives factorizability; the bounds use only that the identity and the witness have different labels). Suppose that the table has a triangle for every generator inverse and for every relator prefix merge from a prefix other than the identity, that the relators have length at most $\ell_* \geq 2$, and that there is a singleton for every label. If e_0 is a relator tolerance, then*

$$\Delta_J(\Phi) \geq \frac{e_0^2}{4096 \ell_*^2 d}, \quad \Delta_\diamond(\Phi) \geq \frac{e_0^2}{4096 \ell_*^2}.$$

Proof. We follow [16, Appendix I]. Let Ψ be a noisy operation with unitary U as in (1) and bath blocks U_{ab} , write p_g for the singleton position of the label g , and put $X_g = U_{p_g p_g}$. In the Choi case let $\delta_J = d_J(\Phi, \Psi)$ and $k = \sqrt{d\delta_J}$; in the diamond case let $\eta = d_\diamond(\Phi, \Psi)$ and $k = \sqrt{\eta}$. Suppose the distance is at most the bound claimed, so that $64\ell_*k \leq e_0 \leq 1$.

Singletons. Row p_g of every Kraus matrix vanishes off the diagonal, so the projection onto the vectors $|p_g, b\rangle$ with $b \neq p_g$ has target weight 0 and comparison weight $\tau(I - X_g X_g^*)/d$; hence $\tau(I - X_g X_g^*) \leq k^2$. In the diamond case the input $|p_g\rangle$ leaves p_g with probability $\tau(I - X_g^* X_g) \leq k^2$. Since $(1 - s)^2 \leq 1 - s^2$ on $[0, 1]$, the polar part of X_g , completed to a unitary v_g , satisfies $\|X_g - v_g\|_2 \leq k$.

Entries. Let the triangle entry at position (i_0, j_0) carry the coefficient $c \in \{\frac{3}{5}, \frac{4}{5}, -\frac{3}{5}\}$ and the label g . The target Choi state annihilates $|i_0, j_0\rangle - c|p_g, p_g\rangle$, so the normalized projection onto this vector gives $\|U_{i_0 j_0} - cX_g\|_2 \leq \sqrt{1 + c^2}k$. In the diamond case the input $(|j_0\rangle|0\rangle + |p_g\rangle|1\rangle)/\sqrt{2}$ gives $\|U_{i_0 j_0} - cX_g\|_2 \leq \sqrt{2(1 + c^2)}k$. In both cases $\|U_{i_0 j_0} - cv_g\|_2 < 3k$.

Triangles. For a triangle (x, y, z) on positions $u, u + 1$, rows u and $u + 1$ of U are orthogonal. Their entries outside columns $u, u + 1$ have target weight 0, so the product of these two outside parts has $\|\cdot\|_2 \leq k$; in the diamond case it is at most $\sqrt{2}k$, by comparing the leakage into and out of $\{u, u + 1\}$. Replacing the four block entries by av_1, bv_y, bv_x and $-av_z$ costs at most $12k$, so $ab\|v_1 v_x^* - v_y v_z^*\|_2 \leq 13k$, or $(12 + \sqrt{2})k$ in the diamond case. With $w_g = v_1^* v_g$ and $ab = \frac{12}{25}$ this gives $\|w_x w_y - w_z\|_2 < 32k$.

Relators and witness. Assign $s \mapsto w_s$ to the generators, so that $w_1 = I$. A relator of length ℓ has at most $\ell - 1$ merges, each costing below $32k$ and nothing from an identity prefix, and replacing the label of each inverse letter by the adjoint of its generator costs below $32k$, by its inverse triangle. So every relator defect is at most $64\ell_*k \leq e_0$, the tolerance gives $\|w_j - I\|_2 \leq \frac{1}{2}$, and therefore $\Re\tau(w_j) \geq \frac{7}{8}$. On the other hand the target has no coherence between the identity and witness singletons, while $\langle p_1, p_1 | J(\Psi) | p_j, p_j \rangle = \tau(X_1 X_j^*)/d$. Since $J(\Psi) - J(\Phi)$ is traceless with trace norm $2\delta_J$, its operator norm is at most δ_J , so $|\tau(X_1 X_j^*)| \leq k^2$ and $|\tau(w_j)| \leq k^2 + 2k$. In the diamond case the input $(|p_1\rangle|0\rangle + |p_j\rangle|1\rangle)/\sqrt{2}$ gives $|\tau(w_j)| \leq 2k^2 + 2k$. As $k \leq 1/128$, both contradict $\Re\tau(w_j) \geq \frac{7}{8}$. So every noisy operation lies farther from Φ than the bounds, and every point of $\mathcal{K}_d$ lies at least that far. $\square$

Corollary 7.4 (Theorem III, fully referenced part). $\Delta_J(\Phi_{\text{full}}) \geq \delta_5^2/65188028416 > 2^{-(2\mathfrak{J}+36)}$ and $\Delta_\diamond(\Phi_{\text{full}}) \geq \delta_5^2/1478656 > 2^{-(2\mathfrak{J}+21)}$.

Proof. Theorem D.4 shows that $e_0 = \delta_5$ is a relator tolerance: relator defects at most δ_5 force $\|v_j - I\|_2 < 2^{-17} \leq \frac{1}{2}$. Lemma 7.3 with $\ell_* = 19$ and $d = 44086$ gives these bounds, since $4096 \cdot 19^2 = 1478656$ and $1478656 \cdot 44086 = 65188028416$. $\square$

Remark 7.5 (Thirteen qubits). Apply the argument of Lemma 7.3 to Wang and Zhi's relation system with $\ell_* = 6$ and $e_0 = 2^{-(\mathfrak{J}+10)}$ (Proposition 4.1), using the singletons of the signals p, q in place of those of the identity and the witness. The fully referenced channel has $d = 2 \cdot 1866 + 949 = 4681 > 2^{12}$ positions. Its bounds are $\Delta_J \geq e_0^2/(4096 \cdot 36 \cdot d) > 2^{-(2\mathfrak{J}+50)}$ and $\Delta_\diamond \geq e_0^2/(4096 \cdot 36) > 2^{-(2\mathfrak{J}+38)}$. We do not use this.

The table, deletion and channel facts listed at the start of this section are checked on the released data of [16] by `verify_earlier_channels.py`; the facts about the model Γ (the relations hold, $j \neq 1$, the labels are distinct) are checked by the two scripts in `earlier/model/`, adapted from that report's package [16, Appendix L]. Repeated use of these channels obeys (4) with the gaps above, and for Φ_2 and Φ_{15} the flat-probe budget applies with the rank-two probe on the identity and witness positions.

8 Verification

The scripts and data files accompany this paper as ancillary files. The verification scripts use only the Python standard library (`build_f5_root_certificate.py`, which produced `f5_root_sos.json`, uses `sympy` and is not part of the verification); running `python verify_all.py` from their directory runs each of the following and ends with `OK ALL`.

- `relation_compiler.py` rebuilds the relation system of Appendix A, checks every relation and every prefix triangle in the amalgam model, adds the inverse triangles from the inverse elements, checks that the 949 labels are distinct, and builds the triangle table.
- `build_max_anchor_wz_channel.py` rebuilds Φ_{12} from that table in a scratch directory, and `verify_all.py` checks that its decompressed content equals the shipped payload's.
- `verify_payload_table.py` imports the compiler and checks that the payload's triangle table is the compiler's, that every block carries the Kraus pattern of its triangle, that the identity and all 88 letter labels have singletons, and that the signals are the compiler's p and q .
- `verify_channel_payload.py` checks the payload without the compiler: its hash and counts, disjoint supports, exact normalization over the denominator 5, distinct singleton labels with their positions, and the signal positions.
- `verify_compiler_constants.py` and `verify_original15_leakage_constants.py` recompute the constants of Sections 3, 4 and 7 from $a = 3/5$ and $b = 4/5$, the second together with the constants of Lemma 7.3 and, from the shipped data, the witness margin for the reduced channels; `verify_two_test_certificate.py` computes the target values of the witness from the payload.
- `verify_earlier_channels.py` checks, on the files of [16] shipped in `earlier/`, the table, deletion and channel facts listed at the start of Section 7: that Φ_{full} has the SHA-256 recorded with [16], its dimension, its block pattern and one singleton per label, that the relators are distinct and freely reduced and that the witness generator has label index 85, that no proper prefix of a relator is the identity, that the table is exactly the prefix and inverse triangles, the relator walks, the 15990 retained triangles, both deletion families, with both orientations of the inverse triangles in family (a), label coverage, and for each of Φ_1 , Φ_2 , Φ_{15} the singletons, the Kraus pattern and the exact normalization.
- `earlier/model/check_model.py` and `earlier/model/check_labels.py`, adapted from the package of [16], check that every relator of the 43-generator presentation holds in D_0 , that the image of j is the reduced word of Section 7, and that the 11230 labels of Φ_{full} are distinct elements of Γ satisfying all its triangles.
- `verify_all.py` also compares the data files with `SHA256SUMS.txt`.
- The three scripts
`verify_nt_certificate.py`,
`verify_rigidity_constants.py`,
`verify_f5_old15_rigidity.py`
check the spectral certificates of Appendices B and D, the transfer constants, Wang and Zhi's exponent comparison $8M_0 < \mathfrak{J}$, which implies the one in Lemma C.1, and the facts about the 43-generator presentation used in Appendix D: its relator families, the twelve Steinberg

relators, the 144 forward compression relators with their targets of length 1 or 16, identical in both copies, and the witness word.

The scripts check finite identities only. The analytic steps are proved in the text, and Wang and Zhi's theorems are used as stated in Appendix C. The cell count of Lemma D.2 is not checked by a script; Lemma C.1 tolerates any constant up to 2^{M_0} , far beyond any slip in it. That C enters Wang and Zhi's recipe only at its last step, which Lemma C.1 uses, can be read off from [28, (3.21)–(3.22)], where C appears only in $\varepsilon_{\text{in}} = \min\{1, \delta_*/(C+1)\}$.

9 Discussion

A usable gap. The constants are explicit but astronomically small, because the tolerance of the reflection estimate is. Wang and Zhi's polynomial needs the penalty weight $\mu = 2^{23+30}$ [28, (2.20)] for the same reason. A rigidity theorem with a larger tolerance, for a smaller relation system, would improve both their weight and our gaps.

Smaller channels. Twelve qubits are forced only for the present table, whose 1866 disjoint triangles alone need $3732 > 2^{11}$ positions. Shared block coordinates or a smaller rigid relation system could reduce the dimension and the Choi rank 949. A diamond bound independent of the dimension for compiled channels probably follows from a local version of the coherence test; we do not pursue it here.

A compiler dictionary. Theorem 3.4 applies to any quantitative nonhyperlinearity witness with short relators. A systematic dictionary between rigid relation systems, explicit channel witnesses and repeated-use lower bounds may be more useful than any one example. Musat and Rørdam's correspondence between the closure and hyperlinear traces [20] is the qualitative form of such a dictionary, and Slofstra's embedding theorem [25] plays this role for nonlocal games. An explicit correlation matrix of unitaries outside the closure of the matrix ones would likewise give an explicit Schur channel outside $\mathcal{K}_n$ [18, Proposition 3.9 and Corollary 3.11].

Use of generative AI

Generative AI systems were used during mathematical exploration, code development, checking and editorial revision. The authors selected the arguments, checked the proofs and the finite certificates, and take responsibility for the mathematical claims and the manuscript.

A The relation system

The relation system is Wang and Zhi's auxiliary system [28, Sections 2 and 5.1], with the identification described below; their group pair is a Kun–Thom pair [14, Theorem E]. Let $\mathbb{F}_8 = \mathbb{F}_2[\alpha]/(\alpha^3 + \alpha + 1)$, let ξ_1, ξ_2, ξ_3 be variables, and put

$$H = \text{EL}_3(\mathbb{F}_8[\xi_1, \xi_2, \xi_3]), \quad G = \text{EL}_3(\mathbb{F}_8[\xi_1^{\pm 1}, \xi_2^{\pm 1}, \xi_3^{\pm 1}]) \rtimes \text{SL}_3(\mathbb{Z}), \quad \Lambda = G *_H G,$$

where $A \in \text{SL}_3(\mathbb{Z})$ acts on Laurent monomials by $\xi^v \mapsto \xi^{Av}$ for exponent vectors v . Let $I = \{0, 1, 2\}$, $L = \{0, 1, 2, 3\}$, $D = \{1, 2, 3\}$ and $\xi_0 = 1$. The 72 root symbols are

$$x_r, y_r, c_r, z_{\ell,r}, d_{\ell,r}, e_{\ell,r}, p_{\{a,b\},r} \quad (r \in I, \ell \in L, 1 \leq a < b \leq 3), \quad c_k, d_{\ell,k}, e_{\ell,k} \quad (k = 3, 4).$$

Write $\widehat{c}_k = c_k$, $\widehat{d}_{\ell,k} = d_{\ell,k}$, $\widehat{e}_{\ell,k} = e_{\ell,k}$ for $0 \leq k \leq 4$. There are six action symbols U_{ab} for ordered $a \neq b$ in D , and a second-copy symbol t . The 843 balanced equations are:

- (R1) *Root commutations*, 585 equations, with all indices $r, s \in I$ and $\ell, m \in L$; the symbols with $k = 3, 4$ and the p symbols occur in no commutation relation. Each of the six families x, y, c, z, d, e commutes internally; every c_r commutes with every x_s, y_s ; every $d_{\ell,r}$ commutes with every x_s and every $z_{m,s}$; every $e_{\ell,r}$ commutes with every y_s and every $z_{m,s}$.
- (R2) *Root crossings*, 81 equations. For $i, j \in I$ and $\ell \in L$, with $i + j$ the integer sum in $\{0, \dots, 4\}$,
$$x_i y_j = \widehat{c}_{i+j} y_j x_i, \quad x_i z_{\ell,j} = \widehat{d}_{\ell,i+j} z_{\ell,j} x_i, \quad y_i z_{\ell,j} = \widehat{e}_{\ell,i+j} z_{\ell,j} y_i.$$
- (R3) *Integer action*, 15 equations. For pairwise distinct $i, j, k \in D$: $U_{ij} U_{ik} = U_{ik} U_{ij}$ and $U_{ji} U_{ki} = U_{ki} U_{ji}$ for $j < k$ (three each); $U_{ij} U_{jk} = U_{ik} U_{kj} U_{ij}$ (six); and $U_{ij} U_{ji}^{-1} U_{ij} = U_{ji}^{-1} U_{ij} U_{ji}^{-1}$ for $i < j$ (three).
- (R4) *Product roots*, 18 equations. For ordered $a \neq b$ and $r \in I$: $d_{a,r} e_{b,0} = p_{\{a,b\},r} e_{b,0} d_{a,r}$.
- (R5) *Common compression*, 126 equations. Put $z_{\ell,r}^{ab} = p_{\{a,b\},r}$ if $\ell = b$ and $z_{\ell,r}^{ab} = z_{\ell,r}$ otherwise. For ordered $a \neq b$, $r \in I$ and $\ell \in L$: $U_{ab} x_r = x_r U_{ab}$, $U_{ab} y_r = y_r U_{ab}$, $U_{ab} z_{\ell,r} = z_{\ell,r}^{ab} U_{ab}$; and the same 18 equations for t with $(a, b) = (1, 2)$.
- (R6) *Auxiliary additions*, 18 equations. For $A \in \{c, d_0, d_1, d_2, d_3, e_0, e_1, e_2, e_3\}$: $\widehat{A}_3 = \widehat{A}_0 \widehat{A}_1$ and $\widehat{A}_4 = \widehat{A}_1 \widehat{A}_2$.

Every root symbol is an involution (72 equations). Finally add the signals p, q with $p = U_{12}^{-1} h U_{12}$ and $q = t^{-1} h t$, where $h = z_{2,0}$. This gives $917 = 843 + 72 + 2$ relators, each of length at most six. Wang and Zhi's auxiliary system has 18 ordered symbols $p_{ab,r}$; we use one symbol for $p_{ab,r}$ and $p_{ba,r}$, which leaves 72 root symbols and the same 843 equations.

In the model, with $e_{kl}(z) = I + z E_{kl}$, x_r, y_r, c_r are the elementary matrices $e_{12}(\alpha^r)$, $e_{23}(\alpha^r)$, $e_{13}(\alpha^r)$; $z_{\ell,r}, d_{\ell,r}, e_{\ell,r}$ are $e_{31}(\alpha^r \xi_\ell)$, $e_{32}(\alpha^r \xi_\ell)$, $e_{21}(\alpha^r \xi_\ell)$; $p_{\{a,b\},r} = e_{31}(\alpha^r \xi_a \xi_b)$; the roots with $k = 3, 4$ use α^3 and α^4 ; U_{ab} is the shear $I + E_{ab}$ in the first factor and t the same shear in the second. Elementary matrix identities verify (R1), (R2), (R4) and (R6), and the shear action verifies (R3) and (R5). The signals have trivial integer action and coefficient $\xi_1^{-1} \xi_2$, which is not a polynomial, so they lie outside H , in different factors of the amalgam, so $p \neq q$ by the normal form for amalgamated products [24], as in [28, (2.9) and Lemma 2.2], and $\tau(pq^*) = 0$.

Proposition A.1 (Finite data). *The relation system has 917 relators of length at most six and 81 generators. In Λ every relator holds, the 949 labels are distinct, every prefix and inverse triangle holds, the 88 letter labels are distinct, and the signals have label indices 87 and 88. There are 1866 triangles.*

Proof. This is checked by `relation_compiler.py` in exact arithmetic: Laurent polynomials over $\mathbb{F}_8$ as finite dictionaries, action matrices over $\mathbb{Z}$. Two prefixes are identified only when they are equal in one factor, or when a prefix with trivial action equals an explicitly computed element of H , which certifies membership in H . Non-membership in H is certified by a nontrivial integer action or, for trivial action, by a negative exponent in some entry. No numerical linear algebra is used. $\square$

B Spectral certificates

For unitaries $u_1, \dots, u_n$, the *conjugation Laplacian* is $\mathcal{L} = I - T$ with $T(x) = \frac{x}{2} + \frac{1}{4n} \sum_j (u_j x u_j^* + u_j^* x u_j)$, an operator on $L^2(M_d, \tau)$.

The finite-field root certificate

The root input is Wang and Zhi's [28, Propositions 3.1 and 3.2]. Theorem 4.2 uses their Theorem 5.2 as a whole, so this subsection does not enter our proofs; we record it for completeness, in the form our scripts check. The Netzer–Thom certificate of the next subsection does enter, through (13) in Appendix D. Let $U_1 = e_{12}(\mathbb{F}_8)$, $U_2 = e_{23}(\mathbb{F}_8)$ and $U_3 = e_{31}(V)$ with $V = \mathbb{F}_8 \oplus \mathbb{F}_8 \xi_1 \oplus \mathbb{F}_8 \xi_2 \oplus \mathbb{F}_8 \xi_3$. Put $P_i = |U_i|^{-1} \sum_{g \in U_i} g$, $\Delta_H = \sum_i (1 - P_i)$ and $D_H = \Delta_H/6$. For each pair, $2 - P_i - P_j$ has spectrum in $\{0, 1, 2, 1 \pm 1/\sqrt{8}\}$ [28, (3.2)], by the root-subgroup method of Ershov and Jaikin-Zapirain [7]. With

$$E_1(t) = t(t-2)(8(t-1)^2 - 1), \quad E_2(t) = \frac{1}{14}t(t-1)(8(t-1)^2 - 1),$$

$$E_*(t) = -\frac{64}{7}t(t-2)(t-1)^2, \quad V_*(t) = (t - \frac{21}{32})E_*(t),$$

one has, modulo $m(t) = t(t-1)(t-2)((t-1)^2 - 1/8)$,

$$t^2 - \frac{5}{8}t = \frac{3}{8}E_1(t)^2 + \frac{11}{4}E_2(t)^2 + 2V_*(t)^2 + \frac{7}{512}E_*(t)^2,$$

and summing over the three pairs expresses $D_H^2 - D_H/24$ as a sum of twelve Hermitian squares with positive rational weights. For the conjugation Laplacian $\mathcal{L}_H$ of the averaging list of length 12288 (each element of U_1 and U_2 repeated 512 times, each element of U_3 once), a unitary assignment whose 801 Heisenberg tests have defect at most η satisfies [28, Proposition 3.2]

$$\langle (\mathcal{L}_H^2 - \mathcal{L}_H/24)y, y \rangle \geq -2^{64}\eta \quad (\|y\| \leq 1). \quad (8)$$

An independently computed Netzer–Thom certificate

Wang and Zhi's integer input uses the property (T) certificate of Netzer and Thom [21] and its transfer to the fifteen balanced equations (R3) [28, Lemma 3.3]. We recomputed both independently. Let $v_{ij} = I + E_{ij} \in \text{SL}_3(\mathbb{Z})$ and $\Delta_K = 12 - \sum_{i \neq j} (v_{ij} + v_{ij}^{-1})$. The ball of word radius two has 121 elements $a_1, \dots, a_{121}$. The data file contains an integer matrix $S \in M_{120,121}(\mathbb{Z})$ with zero row sums; with $P = 10^{-14}S^T S$,

$$\varepsilon_{NT} = \frac{561}{2000}, \quad c = (\Delta_K^2 - \varepsilon_{NT}\Delta_K) - (a_1^*, \dots, a_{121}^*)P(a_1, \dots, a_{121})^T, \quad \|c\|_1 = \frac{572700482439}{25000000000000}.$$

By the correction lemma of Netzer and Thom [21, Lemma 2.1], $c + 4\|c\|_1\Delta_K$ is a sum of Hermitian squares. The lemma applies because c is Hermitian, has augmentation zero and is supported in the ball of radius four. Its constant is 4, not 8, because no generator is its own inverse. We apply the lemma in the group algebra of the free group on the six generators, to the Hermitian part $(\tilde{c} + \tilde{c}^*)/2$ of the lift $\tilde{c}$ of c used by the data, which has the same support radius and at most the same ℓ^1 -norm; since the quadratic forms involved are real, this suffices, and the resulting inequality holds for arbitrary unitaries before any relation is imposed. Hence $\Delta_K^2 - \Delta_K/6$ is a sum of Hermitian squares up to the comparison residual handled next, since $\varepsilon_{NT} - 4\|c\|_1 = \frac{1180424517561}{6250000000000} > \frac{1}{6}$.

For the transfer, the residual has 6722 coefficient comparisons, which reduce under cyclic conjugation and inversion to 858 canonical words. The file `nt_word_reductions.json.gz` gives for each a sequence of at most 14 replacements by the fifteen balanced relations, each checked symbolically. To bound it, lift c word by word to the free group on the six unitaries, group its coefficients z_u by the value of the word u in $\text{SL}_3(\mathbb{Z})$, and let u_C be the least word of each class C , by length and then lexicographically. The coefficient ledger, as in [28, (3.10)], is $C_K = 2 \cdot 24^{-2} \sum_C \sum_{u \in C, u \neq u_C} |z_u| = \frac{37610695003407823}{2880000000000000}$; the factor 2 pays for conjugation and 24^{-2} converts Δ_K to $\mathcal{L}_K$. Each comparison word $u_C^{-1}u$ reduces to the empty word in at most 14 replacements, so

the residual constant is $14C_K$, and six unitaries whose fifteen balanced equations have defect at most η satisfy

$$\langle (\mathcal{L}_K^2 - \mathcal{L}_K/144)y, y \rangle \geq -\frac{263274865023854761}{1440000000000000} \eta > -19\eta \quad (\|y\| \leq 1), \quad (9)$$

where $\mathcal{L}_K = \Delta_K/24$ is the conjugation Laplacian of the six unitaries. Wang and Zhi's version has constant 26 in the same normalization [28, (3.12)]. Either serves, by Lemma C.1.

C Statements used from Wang and Zhi

We use three results of Wang and Zhi as theorems, proved in [28], where several steps follow the cited proofs of Alekseev, Liu and Thom and of Thom, and one extension of a fourth. They make explicit the spectral correction and reconstruction of Alekseev, Liu and Thom [1, Theorems 4.3 and 5.2, Corollary 5.1] and the normalization argument of Thom [27, Proposition 3.1, Lemma 4.1, Theorem 4.2]. Below, $u_1, \dots, u_n \in U(d)$, and $T, \mathcal{L}$ are as in Appendix B, with $\mathcal{E}(x) = \langle \mathcal{L}x, x \rangle = \frac{1}{4n} \sum_j \|[u_j, x]\|_2^2$. For a unital $*$ -subalgebra $\mathcal{A}$, $E_{\mathcal{A}}$ is the τ -preserving conditional expectation onto $\mathcal{A}$.

The tolerance recipe. For an integer $n \geq 1$ and rationals $0 < \lambda < 1$, $C \geq 0$, $0 < \theta < 1$, Wang and Zhi define a rational $\varepsilon_{\text{in}}(n, \lambda, C, \theta) > 0$ by an explicit procedure [28, Section 3.5, (3.21)–(3.22)]. Its last step is $\varepsilon_{\text{in}} = \min\{1, \delta_*/(C+1)\}$, where δ_* depends on n, λ, θ only. Our n is their h , and the input C is not their constant C_0 .

Lemma C.1 (extending [28, Lemma 5.1]). *Let $\mathfrak{J} = 2^{2^{1122000}}$, $\delta = 2^{-\mathfrak{J}}$, $\theta = 2^{-140096}$ and $M_0 = 2^{2^{1121215}}$. If $n \leq 2^{14}$, $2^{-9} \leq \lambda/2 \leq 1/4$ and $C \leq 2^{M_0}$, then $\delta \leq \varepsilon_{\text{in}}(n, \lambda, C, \theta)$.*

Proof. Wang and Zhi state the inequality for their two inputs. In the proof of their Lemma 5.1 they show $\delta_* \geq 2^{-4M_0}$, using only $n \leq 2^{14}$, $2^{-9} \leq \lambda/2 \leq 1/4$ and $\theta = 2^{-140096}$; their bound $C \leq 2^{64}$ enters only in the next step, through $C+1 \leq 2^{65}$. Here $C+1 \leq 2^{M_0+1}$, so $\varepsilon_{\text{in}} \geq 2^{-4M_0}/2^{M_0+1} \geq 2^{-6M_0}$, and $6M_0 < 8M_0 < \mathfrak{J}$. The script `verify_rigidity_constants.py` repeats their comparison $8M_0 < \mathfrak{J}$. $\square$

Theorem C.2 ([28, Corollary 4.3], a quantitative form of [1, Theorems 4.3 and 5.2]). *Let $0 < z < 1$ be rational and $\theta = z^8/2^{96}$. If $\langle (\mathcal{L}^2 - \lambda\mathcal{L})y, y \rangle \geq -C\varepsilon$ for all $\|y\| \leq 1$, with $0 \leq \varepsilon \leq \varepsilon_{\text{in}}(n, \lambda, C, \theta)$, there is a unital $*$ -subalgebra $\mathcal{A} \subseteq M_d$ with*

$$\|x - E_{\mathcal{A}}x\|_2 \leq (\lambda/2)^{-1/2} \mathcal{E}(x)^{1/2} + 3z \quad (\|x\| \leq 1), \quad \max_j \|[u_j, a]\|_2 \leq 6\sqrt{n}z \quad (a \in \mathcal{A}, \|a\| \leq 1).$$

For unital $*$ -subalgebras $\mathcal{A}, \mathcal{B}$ write $\mathcal{B} \subseteq_{\eta} \mathcal{A}$ if every contraction of $\mathcal{B}$ is within η in $\|\cdot\|_2$ of a contraction of $\mathcal{A}$.

Theorem C.3 ([28, Theorem 4.1], a quantitative form of [27, Theorem 4.2]). *Let $k_W = 2^{-9}$, $r_W = 2^{-17335}$ and $\eta_W = 2^{-17313}$. Let $\mathcal{A}, \mathcal{B} \subseteq M_d$ be unital $*$ -subalgebras and $V_1, \dots, V_m \in U(d)$, with $\mathcal{E}_V(x) = (4m)^{-1} \sum_j \|[V_j, x]\|_2^2$. If*

$$\|x - E_{\mathcal{B}}x\|_2 \leq k_W^{-1/2} \mathcal{E}_V(x)^{1/2} + r_W \quad (\|x\| \leq 1), \quad \max_j \|[V_j, b]\|_2 \leq r_W \quad (b \in \mathcal{B}, \|b\| \leq 1),$$

and $V_j^ \mathcal{A} V_j \subseteq_{\eta_W} \mathcal{A}$ for every j , then $V_j \mathcal{A} V_j^* \subseteq_{2^{-20}} \mathcal{A}$ for every j , uniformly in d and m .*

Theorem C.4 ([28, Theorem 5.2]). *Suppose the 81 root symbols of Wang and Zhi's auxiliary system are represented by symmetries and the seven integer symbols by unitaries in M_d . If every one of the 843 balanced equations has defect less than $\delta/128$, then $\|U_{12}^* h U_{12} - t^* h t\|_2 \leq 5 \cdot 2^{-20}$, where $h = z_{2,0}$.*

Proposition 4.1 applies it to our 72-symbol system by repeating the identified symbols.

D Characteristic-five rigidity for the 43-generator presentation

This appendix gives the tolerance for the 43-generator presentation of Section 7. Its positive root subgroup is generated by three families a_r, b_r, c_r , $r \in \{0, 1, 2, 3\}$, of exponent five; each family is commutative, and for the consecutive pairs (a, b) , (b, c) , (c, a) the commutators $[x, y] = xyx^{-1}y^{-1}$ are central in the pair, by (5). The integer part has six action generators t_{ij} in each of two copies. The root certificate is new, but its method is Wang and Zhi's [28, Propositions 3.1 and 3.2]: a sum of squares modulo the minimal polynomial of each pair, summed over the three pairs, then transferred to approximate relations by a collection bound, as in [28, (3.7)]. The pair spectrum (Lemma D.1) comes from the method of [7] and refines the cosine bound $1/\sqrt{5}$ of [16, Lemma B.1]; the certificate (11) and the cell count of Lemma D.2 are ours. Theorem D.4 follows the proof of [28, Theorem 5.2], which makes Thom's proof of [27, Theorem 1.3] quantitative, using Theorems C.2 and C.3.

The pair spectrum and its certificate

Fix one consecutive pair, written $X = \langle x_1, \dots, x_4 \rangle$ and $Y = \langle y_1, \dots, y_4 \rangle$, and let $\mathcal{G}_2$ be the universal group with relations $x_i^5 = y_j^5 = 1$, $[x_i, x_{i'}] = [y_j, y_{j'}] = 1$, and $c_{ij} = [x_i, y_j]$ commuting with every x_u and y_u . It is the class-two group $\mathbb{F}_5^4 \times \mathbb{F}_5^4 \times \mathbb{F}_5^{16}$ with multiplication $(a, b, e)(a', b', e') = (a + a', b + b', e + e' - a' \otimes b)$, where $(a' \otimes b)_{ij} = a'_i b_j$ (this is $y_j x_i = c_{ij}^{-1} x_i y_j$), of order 5^{24} , so every element has a unique collected form

$$x_1^{a_1} \cdots x_4^{a_4} y_1^{b_1} \cdots y_4^{b_4} \prod_{(i,j)} c_{ij}^{e_{ij}}, \quad a_i, b_j, e_{ij} \in \{0, \dots, 4\}, \quad (10)$$

with the product in a fixed order of the sixteen pairs. Let P_X, P_Y be the averages over the two elementary abelian subgroups in the group algebra and $\Xi = 2 - P_X - P_Y$.

Lemma D.1 (Pair spectrum). *In every unitary representation of $\mathcal{G}_2$, the spectrum of Ξ lies in $\{0, 1, 2\} \cup \{1 \pm 5^{-s/2} : 1 \leq s \leq 4\}$.*

Proof. This is the root-angle computation of Ershov and Jaikin-Zapirain [7], as in [28, (3.2)]. In an irreducible representation the central c_{ij} act by scalars and define a bicharacter $X \times Y \rightarrow \mu_5$; let s be its rank over $\mathbb{F}_5$. If $s = 0$, X and Y act through a character and $P_X, P_Y \in \{0, 1\}$. If $s > 0$, the representation factors through a Heisenberg representation of degree 5^s on which the X - and Y -fixed spaces, when nonzero, are lines whose unit vectors have squared overlap 5^{-s} . Two projections with this angle contribute the eigenvalues $1 \pm 5^{-s/2}$ to Ξ ; the rest of the space contributes 1 or 2. $\square$

Let

$$m_5(t) = t(t-1)(t-2)(t-\frac{6}{5})(t-\frac{4}{5})(t-\frac{26}{25})(t-\frac{24}{25})((t-1)^2 - \frac{1}{5})((t-1)^2 - \frac{1}{125})$$

and $q_5(t) = t^2 - \frac{11}{20}t$. The file `f5_root_sos.json` contains 23 rational polynomials s_ν and a rational polynomial h_5 with

$$q_5(t) = \sum_{\nu=1}^{23} s_\nu(t)^2 + h_5(t)m_5(t), \quad (11)$$

verified coefficient by coefficient in exact arithmetic. Since $\mathcal{G}_2$ is finite, Lemma D.1 applied to the regular representation gives $m_5(\Xi) = 0$ in the group algebra, and hence $\Xi^2 - \frac{11}{20}\Xi \geq 0$ in every representation.

Approximate relations

Let unitaries be assigned to the twelve positive root generators so that every fifth-power, within-family commutation and class-two relation of the pair has defect at most e . For $g \in \mathcal{G}_2$ let W_g be the unitary evaluation of its collected word (10).

Lemma D.2 (Collection). $\|W_g W_h - W_{gh}\|_2 \leq 2^{19}e$ for all $g, h \in \mathcal{G}_2$.

Proof. Rewrite the word of $W_g W_h$ into the collected word of gh by free identities and *cells*, each the replacement of a subword by an equivalent one using one relation or its inverse. A free identity does not change the unitary evaluation, and a cell changes it by at most e in $\|\cdot\|_2$, by unitary invariance. Write the product as $x^{\mathbf{a}} y^{\mathbf{b}} Z x^{\mathbf{a}'} y^{\mathbf{b}'} Z'$, where $x^{\mathbf{a}}$ and $y^{\mathbf{b}}$ have at most 16 letters each and Z, Z' consist of at most 64 commutator blocks c_{ij} each. Process the word in the following order.

- (1) Move the ≤ 16 letters of $x^{\mathbf{a}'}$ left across the ≤ 64 blocks of Z : at most 1024 class-two cells.
- (2) Move them, one at a time and leftmost first, across the ≤ 16 letters of $y^{\mathbf{b}}$ using the free identity $y_j x_i = c_{ij}^{-1} x_i y_j$. This creates at most 256 blocks c_{ij}^{-1} . Each is moved right, at once, across the at most 16 letters of $y^{\mathbf{b}}$ and 16 letters of $x^{\mathbf{a}'}$ still to its right, to join Z : at most 8192 cells.
- (3) Move the ≤ 16 letters of $y^{\mathbf{b}'}$ left across the ≤ 320 blocks now between them and $y^{\mathbf{b}}$: at most 5120 cells.
- (4) Sort the ≤ 32 x -letters and the ≤ 32 y -letters with within-family commutations, and reduce exponents above four with fifth powers: at most $2(496 + 4) = 1000$ cells.
- (5) The central zone has at most 384 blocks $c_{ij}^{\pm 1}$. Sort them into the fixed order; exchanging two adjacent blocks costs four class-two cells, and at most $\binom{384}{2}$ exchanges are needed: at most 294144 cells.
- (6) Cancel $c_{ij} c_{ij}^{-1}$ freely. The net exponent of each pair lies in $[-16, 8]$, so at most four reductions by $c_{ij}^{\pm 5} = 1$ per pair are needed. The free identity $[x^5, y] = \prod_{k=4}^0 x^k c x^{-k}$, with $c = [x, y]$, and at most 10 class-two cells give $c^5 = [x^5, y]$; two fifth-power cells finish it: at most $16 \cdot 4 \cdot 12 = 768$ cells.

The total is at most $310248 < 2^{19}$ cells. □

Let $N = 5^4$ and, for each of the three families i , let $\tilde{P}_i = \frac{1}{2N} \sum_g (\text{Ad } W_g + \text{Ad } W_g^*)$, the sum over the N collected words of that family. This operator on $L^2(M_d, \tau)$ is selfadjoint, and $D_5 = \frac{1}{6} \sum_i (I - \tilde{P}_i)$ is exactly the conjugation Laplacian of the list of $3N = 1875$ word unitaries.

Proposition D.3 (Root residual). *If all positive-root relations of the three pairs have defect at most e , then for every $\|y\| \leq 1$,*

$$\langle (D_5^2 - D_5/60)y, y \rangle \geq -C_5 e, \quad C_5 = \frac{2^{21}(3C_h(C_m + C'_m) + 9)}{36} < 2^{114}, \quad (12)$$

where $C_m = \sum_k |[t^k]m_5| \max(k-1, 0) 4^k$, $C'_m = \sum_k k |[t^k]m_5| 4^{k-1}$, and C_h bounds $|h_5|$ on $[-4, 4]$. Exactly, $C_5 = 5163853243256234099731107007343755264/403$.

Proof. Let $\mathcal{T}$ be the linear map from the group algebra of $\mathcal{G}_2$ to operators on $L^2(M_d, \tau)$ with $\mathcal{T}(g) = \text{Ad } W_g$, and write $\|S\|_{\infty \rightarrow 2} = \sup\{\|Sy\|_2 : \|y\| \leq 1\}$. Since $\|uyu^* - u'yu'^*\|_2 \leq 2\|u - u'\|_2\|y\|$, Lemma D.2 gives $\|\mathcal{T}(g)\mathcal{T}(h) - \mathcal{T}(gh)\|_{\infty \rightarrow 2} \leq 2^{20}e \leq \mu$, where we put $\mu = 2^{21}e$; conjugation preserves operator-norm contractions.

Fix one pair and put $\tilde{\Xi} = 2 - \tilde{P}_X - \tilde{P}_Y$, a selfadjoint operator on L^2 with spectrum in $[0, 4]$, and $\mathcal{T}(\Xi) = 2 - \mathcal{T}(P_X) - \mathcal{T}(P_Y)$, which maps operator-norm contractions to operators of norm at most 4. Since $\|\Xi\|_{\ell^1} \leq 4$, telescoping gives $\|\mathcal{T}(\Xi)^k - \mathcal{T}(\Xi^k)\|_{\infty \rightarrow 2} \leq (k-1)4^k\mu$, so $\|m_5(\mathcal{T}(\Xi))y\|_2 \leq C_m\mu$ because $m_5(\Xi) = 0$. Also $\|\text{Ad } W_g^* - \text{Ad } W_{g^{-1}}\|_{\infty \rightarrow 2} \leq 2\|W_g W_{g^{-1}} - I\|_2 \leq \mu$, so $\|\tilde{\Xi} - \mathcal{T}(\Xi)\|_{\infty \rightarrow 2} \leq \mu$; expanding $\tilde{\Xi}^k - \mathcal{T}(\Xi)^k = \sum_i \tilde{\Xi}^i (\tilde{\Xi} - \mathcal{T}(\Xi)) \mathcal{T}(\Xi)^{k-1-i}$, with $\|\tilde{\Xi}\|_{2 \rightarrow 2} \leq 4$ on the left and operator norms at most 4 on the right, gives $\|m_5(\tilde{\Xi})y - m_5(\mathcal{T}(\Xi))y\|_2 \leq C'_m\mu$. By (11),

$$\langle (\tilde{\Xi}^2 - \frac{11}{20}\tilde{\Xi})y, y \rangle = \sum_{\nu} \|s_{\nu}(\tilde{\Xi})y\|_2^2 + \langle h_5(\tilde{\Xi})m_5(\tilde{\Xi})y, y \rangle \geq -C_h(C_m + C'_m)\mu.$$

The same bounds give $\|\tilde{P}_i^2 - \tilde{P}_i\|_{\infty \rightarrow 2} \leq 3\mu$. With $\tilde{Q}_i = I - \tilde{P}_i$ and $\Delta = \sum_i \tilde{Q}_i$, the identity $\sum_{i < j} (\tilde{Q}_i + \tilde{Q}_j)^2 = \Delta^2 + \sum_i \tilde{Q}_i^2$ and the three pair inequalities give

$$\langle (\Delta^2 - \Delta/10)y, y \rangle \geq -(3C_h(C_m + C'_m) + 9)\mu.$$

Divide by 36. The script `verify_f5_old15_rigidity.py` evaluates C_m, C'_m, C_h and C_5 exactly. $\square$

The action generators

The action part of the presentation has six commuting and six Steinberg product relations, but not the three relations

$$U_{ij}U_{ji}^{-1}U_{ij} = U_{ji}^{-1}U_{ij}U_{ji}^{-1} \quad (i < j)$$

used in (9), with $U_{ij} = t_{ij}$. The file `st3_opposite_reductions.json` gives three free-group derivations of them, each using 13 cells from the twelve relations, which the verifier replays after checking the twelve relations against `earlier/presentation.json`. So action-relator defect at most e gives balanced defects at most $13e$, and (9) becomes

$$\langle (\mathcal{L}_K^2 - \mathcal{L}_K/144)y, y \rangle \geq -247e \quad (\|y\| \leq 1). \quad (13)$$

The tolerance

Let $\delta_5 = 2^{-3}$ and $z_W = 2^{-17500}$, so $\theta = z_W^8/2^{96} = 2^{-140096}$. Lemma C.1 applies to the root input $(n, \lambda, C) = (1875, 1/60, C_5)$ and to the action input $(6, 1/144, 247)$, so $\delta_5 \leq \varepsilon_{\text{in}}$ for both. Theorem C.2 then gives algebras $\mathcal{A}$ for the root words and $\mathcal{B}$ for the action generators of copy zero, with

$$\|x - E_{\mathcal{A}}x\|_2 \leq k_W^{-1/2}\mathcal{E}(x)^{1/2} + r_W, \quad \max\|[u, a]\|_2 \leq r_W,$$

because $(1/120)^{-1/2}$ and $(1/288)^{-1/2}$ are at most $k_W^{-1/2} = 2^{9/2}$, $3z_W \leq r_W$ and $6\sqrt{1875}z_W \leq r_W$; and likewise for $\mathcal{B}$.

Theorem D.4 (Numerical rigidity of the 43-generator presentation; after Thom [27, Section 5.2] and Wang–Zhi [28, Theorem 5.2]). *Assign unitaries to the 43 positive generators so that each of the 4417 relators has defect at most δ_5 . Then the witness generator satisfies $\|v_j - I\|_2 < 2^{-17}$, in every matrix dimension.*

Proof. We follow the proof of Theorem C.4. Obtain $\mathcal{A}$ and $\mathcal{B}$ as above. For each copy-zero action generator V and each positive root generator g , the compression relation (6) with $\epsilon = 1$ says $VgV^* = w_g$ up to δ_5 , where w_g is g or the word $\beta_f(r, s) = [[f_r, (f^+)_0], [(f^-)_s, f_0]]$ of length 16 in positive roots, with $f^\pm$ as in (6). For a contraction $a \in \mathcal{A}$,

$$\|[V^*aV, g]\|_2 = \|[a, VgV^*]\|_2 \leq 16r_W + 2\delta_5 < 18r_W.$$

Every collected word of a single family has at most 16 letters, so its commutator with V^*aV is below $288r_W$. The Poincaré estimate for $\mathcal{A}$ then gives $V^*\mathcal{A}V \subseteq_{2^{14}r_W} \mathcal{A}$, and $2^{14}r_W \leq \eta_W$. Theorem C.3, with $\mathcal{B}$ and the six copy-zero action generators, gives $V\mathcal{A}V^* \subseteq_{2^{-20}} \mathcal{A}$.

Let t_0, t_1 be the two copies of t_{12} and $u = t_0^{-1}t_1$, Thom's centralizing element [27, Section 5.2]. Their compression relations have the same targets on the twelve shared positive roots, so $\|[u, g]\|_2 \leq 2\delta_5$ for each, and the Poincaré estimate gives $\text{dist}_2(u, \mathcal{A}) < 2^{11}r_W$. Hence $w = t_1t_0^{-1} = t_0ut_0^{-1}$ is within $2^{11}r_W + 2^{-20}$ of a contraction of $\mathcal{A}$, and for the root $h = a_2$,

$$\|hw - wh\|_2 < r_W + 2(2^{11}r_W + 2^{-20}) < 2^{-18}.$$

So $\|[h, w] - I\|_2 < 2^{-18}$, and the naming relator $j = [h, t_1t_0^{-1}]$ has defect at most δ_5 , giving $\|v_j - I\|_2 < 2^{-18} + \delta_5 < 2^{-17}$. $\square$

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